

Hugin-AstA circuitry is a novel central energy sensor that directly regulates sweet sensation in *Drosophila* and mouse

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This **important** work characterizes layers of neuropeptidergic modulations that collectively regulate the intake of sugar in a hunger state-dependent manner. Combinations of genetic, physiological, and behavioral approaches present **convincing** evidence that neurons that release Hugin and Allatostatin A are in an active state in sated flies, leading to suppression of sugar feeding behavior by reducing the sensitivity of sugar-sensitive gustatory neurons that express Gr5a. They also demonstrate that neurons that release Neuromedin U, a vertebrate homolog of Hugin, have common physiological properties as the fly Hugin neurons, revealing a similar function of evolutionarily conserved peptides across animal phyla.

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Abstract

Taste sensation plays a crucial role in shaping feeding behavior and is intricately influenced by internal states like hunger or satiety. Despite the identification of numerous neural substrates regulating feeding behavior, the central neural substrate that linked energy-sensing and taste sensation remained elusive. Here, we identified a novel neural circuitry that could directly sense internal energy state and modulate sweet sensation in the *Drosophila* brain. Specifically, a subset of neuropeptidergic neurons expressing hugin directly detected elevated levels of circulating glucose via glucose transporter Glut1 and ATP-sensitive potassium channel. Upon activation, these neurons released hugin peptide and activated downstream Allatostatin A (AstA)⁺ neurons via its cognate receptor PK2-R1. Subsequently, the activation of AstA⁺ neurons then directly inhibited sweet sensation via AstA peptide and its cognate receptor AstA-R1 expressed in sweet-sensing Gr5a⁺ neurons. We also showed that neuromedin U (NMU), the mammalian homolog of fly hugin, served as an energy sensor to suppress sweet sensation. Therefore, these data identify hugin⁺ neuron as a central energy sensor responsible for regulating sweet sensation across species.

Introduction

Animal behaviors are tightly associated with the internal metabolic states (Munch et al., 2020). When experiencing hunger, animals exhibit behaviors aimed at obtaining food, including elevated sensitivity to appetitive cues and enhanced foraging and feeding behaviors (Augustine et al., 2020; Marella et al., 2012; Yu et al., 2016). In addition, hunger tends to increase anxiety levels and aggressive behaviors (de Rivaz et al., 2022), while inhibiting pain, sleep, and mating behaviors (Alhadeff et al., 2018; Burnett et al., 2016; Sun et al., 2023). These behavioral alterations upon hunger serve to sharpen the focus of starved animals on the tasks related to food acquisition (Deem et al., 2022). Conversely, when animals are sated, food-related behaviors diminish, redirecting their attention to other important behaviors such as reproduction (Burnett et al., 2016; Karigo and Deutsch, 2022). Such adjustments in behaviors help animals to fulfill various internal needs in a fast-changing environment, and also suggests the presence of neural circuitry in the brain that detects and encodes hunger and satiety states to guide individual behavioral choices and maintain metabolic homeostasis.

In mammals, agouti-related peptide (AgRP)- and pro-opiomelanocortin (POMC)-expressing neurons in the hypothalamic arcuate nucleus (ARC) are key sensors of hunger and satiety (Fu et al., 2019). When calories are scarce, AgRP neurons become active, promoting foraging and feeding behaviors while suppressing competing behaviors such as thirst, innate fear, anxiety-related behavior, and social interactions (Burnett et al., 2016; Dietrich et al., 2015). Conversely, satiety signals gradually inhibit the activity of AgRP neurons in response to food or food cues, resulting in reduced appetite (Bruning and Fenselau, 2023; Chen et al., 2015). Antagonistic to AgRP neurons, POMC neurons play a critical role as satiety sensors, suppressing feeding and increasing energy expenditure by releasing a α -melanocyte-stimulating hormone (α -MSH) (Bruning and Fenselau, 2023; Koch et al., 2015). AgRP/POMC neuron activity is regulated by various upstream signals, including hormones like leptin, insulin, and ghrelin, as well as nutrients such as glucose, amino acids, and fatty acids. Leptin, GLP-1, and insulin activate POMC neurons while inhibiting AgRP neurons, reducing food intake (Belgardt et al., 2009; De Solis et al., 2024), whereas ghrelin, secreted during fasting, activates AgRP neurons to promote feeding (Vivot et al., 2023). Furthermore, AgRP and POMC neurons project to various brain regions including paraventricular hypothalamus (PVH), dorsomedial hypothalamus (DMH), ventromedial hypothalamus (VMH), and brainstem, exerting complex regulations on behavior and metabolism (Belgardt et al., 2009; De Solis et al., 2024; Wei et al., 2018). These intertwined circuitry helps to coordinate behaviors and physiological responses in a dynamic environment. Despite the significant progress in the past decades, the direct link of these upstream internal energy sensors, such as AgRP and POMC neurons, to various behavioral outputs, remains incompletely understood. Current research highlights their central role in energy homeostasis but also suggests a complex, multi-level circuitry that modulates these behaviors. Fully elucidating the relative contributions of these neurons and circuits to behavioral and metabolic regulations is an ongoing challenge in the field.

Fruit flies offer an excellent system for studying satiety/hunger sensing and mapping out the complete circuitry towards behavioral consequences, owing to their simple nervous system and comprehensive genetic and neurobiological tools (Lin et al., 2019). Hunger and satiety states modulate a range of behaviors in *Drosophila*, including foraging, feeding, food preference, sleep, and even memory formation (Chouhan et al., 2021; Lin et al., 2019). Hungry flies tend to increase foraging, favor high-calorie food, and extend feeding durations, while satiated flies reduce food intake and shift their food preferences (Lin et al., 2019). Hunger also disrupts sleep, with starved flies becoming more alert to prioritize food-seeking (Willyard, 2008). A constellation of neuromodulators has been shown to steer these behavioral transitions. Dopamine navigates the feeding process by engaging reward pathways upon food detection (Marella et al.,

2012 [\[1\]](#)), whereas serotonin (5-HT) functions to quell feeding behavior during satiety (Yao and Scott, 2022 [\[2\]](#)), simultaneously shaping food preferences and foraging endeavors. Neuropeptide F (NPF), akin to mammalian neuropeptide Y (NPY), is instrumental in instigating feeding bouts during periods of hunger (Malita et al., 2022 [\[3\]](#)). Hugin, bearing a resemblance to mammalian neuropeptide U, influences satiety levels and modulates food intake, regulated by gustatory neurons (Melcher and Pankratz, 2005 [\[4\]](#); Schoofs et al., 2014 [\[5\]](#)). Short neuropeptide F (sNPF) contributes to hunger-induced behaviors by calibrating sensory reactions to food stimuli (Root et al., 2011 [\[6\]](#)). Allatostatin A (AstA) emerges as a satiety signal that, likely through interactions with sweet-sensing neurons, and suppresses feeding activities. (Chen et al., 2016 [\[7\]](#); Hentze et al., 2015 [\[8\]](#); Hergarden et al., 2012 [\[9\]](#)). CCHamide, which is associated with growth modulation, potentially acts as a hunger signal, whereas leucokinin adjusts meal size and manages the physiological aftermath of feeding (Lin et al., 2019 [\[10\]](#)). Corazonin is perceived to be intertwined with the modulation of stress and metabolic responses pertinent to hunger and satiety (Kubrak et al., 2016 [\[11\]](#)). SiFamide is recognized for its role in translating hunger signals into both appetitive and consummatory feeding actions (Martelli et al., 2017 [\[12\]](#)). The intricate interplay among these modulators constitutes a sophisticated regulatory network, the intricacies of which are yet to be delineated comprehensively, underscoring an exciting frontier for forthcoming research aimed at unraveling the neural circuitry that underpins energy homeostasis.

Sugar serves as an important source of satiety signals. The body's precise responses to dietary sugar involve both peripheral organs and the brain, which facilitates the regulation of various behaviors, including sweet perception and food consumption (Oh et al., 2021 [\[13\]](#); Qi et al., 2021 [\[14\]](#); Ugrankar et al., 2018 [\[15\]](#); Zhao et al., 2022 [\[16\]](#)). When exposed to a high-sugar diet, the midgut reacts by secreting Hedgehog (Hh) to suppress sweet sensation (Zhao et al., 2022 [\[16\]](#)). Additionally, hugin⁺ neurons in the ventral nerve cord (VNC) become sugar-sensitive in fed states, leading to the suppression of food consumption via neuropeptide diuretic hormone 44 (DH44)⁺ neurons (Oh et al., 2021 [\[13\]](#)). In the brain, insulin-producing cells (IPCs) and tachykinin (TK)⁺ neurons are also activated in the presence of sugar, contributing to appetite suppression (Hergarden et al., 2012 [\[9\]](#); Lin et al., 2019 [\[10\]](#); Qi et al., 2021 [\[14\]](#)). In contrast to satiety sensing, certain neurons, such as dopaminergic neurons, act as hunger sensors in fruit fly brains, being activated during hunger states and involved in regulation of sleep, sugar memory expression and sweet taste sensation (Marella et al., 2012 [\[17\]](#); Senapati et al., 2019 [\[18\]](#); Yurgel et al., 2019 [\[19\]](#)). However, it remains unclear how these neurons interact to form a regulatory network on energy homeostasis, especially how these neurons link the hunger/satiety state to direct behavioral outputs.

We aimed to dissect out a complete neural pathway that directly sensed internal energy state and modulated food-related behavioral output in the fly brain. More specifically, we focused on the circuitry mediating the effect of satiety to inhibit sweet taste sensation. Among several candidate neuropeptidergic neuron groups, we pinpointed hugin⁺ neurons as a direct satiety sensor in the fly brain. These hugin⁺ neurons were activated by elevated levels of circulating glucose after feeding, which in turn activated downstream AstA⁺ neurons via the interaction of hugin peptide and its cognate receptor PK2-R1. Subsequently, AstA⁺ neurons exerted direct inhibitory effect on sweet-sensing Gr5a⁺ neurons via AstA peptide and its receptor AstA-R1. As a result, sated flies exhibited a reduction in sweet sensitivity and hence food consumption. In summary, we unveiled a novel satiety sensing mechanism in *Drosophila* that detected circulating glucose and regulated sweet perception and feeding. We also showed that NMU, the mammalian homolog of hugin, was also an energy sensor in mouse models involved in regulation of sweet perception, highlighting the conserved regulatory mechanism of central energy sensing and sweet sensation across species.

Results

hugin⁺ neurons were a novel glucose sensor in the fly brain

Flies extend their proboscis in response to appetitive cues, a behavioral element known as the proboscis extension reflex (PER) (Marella et al., 2012 [↗](#); Song et al., 2023 [↗](#)). As previously observed, hunger significantly enhances this reflex (Marella et al., 2012 [↗](#)). In our study, we first replicated this phenomenon: In the state of hunger, flies exhibited elevated PER towards various concentrations of sugar compared to those under fed conditions (**Figure 1A-B** [↗](#)). These results confirmed the notion that sweet sensation was influenced by the internal energy state, elevated by starvation and suppressed by satiety (Inagaki et al., 2014 [↗](#); Marella et al., 2012 [↗](#)). Sweet sensation in flies is mediated by sweet-sensing Gr5a⁺ neurons located on the proboscis (Fujii et al., 2015 [↗](#)). Consistent with the behavioral effect, we also confirmed that Gr5a⁺ neurons exhibited elevated calcium responses to sugar under starvation conditions (**Figure supplementary 1A** [↗](#)).

Dopaminergic neurons in the subesophageal zone (SEZ) region of the fly brain acted as a hunger sensor, promoting sweet sensation via a specific dopamine receptor DopEcR expressed in Gr5a⁺ neurons (Inagaki et al., 2014 [↗](#); Marella et al., 2012 [↗](#)). However, although the gene knockout of tyrosine hydroxylase (TH) (Budnik and White, 1987 [↗](#)), a key enzyme for dopamine biosynthesis, significantly suppressed sweet sensation in starved fruit flies (**Figure supplementary 1B** [↗](#)), these flies still showed a further reduction in sugar-induced PER under sated conditions (**Figure supplementary 1C-D** [↗](#)). These results suggest the existence of dopamine-independent mechanisms for sensing satiety and hunger in fruit flies that exerts modulatory effect on sweet sensation.

In the terminating phase of feeding behavior, satiety signals play a crucial role to cease food consumption and to ensure an appropriate amount of food ingestion. A number of neuropeptidergic neurons were identified as putative satiety signals (Lin et al., 2019 [↗](#)). Notably, some of these neurons, including those secreting neuropeptides LK, TK, hugin, AstA, CCHa1, and CCHa2, are also distributed in the SEZ region (Deng et al., 2019 [↗](#)). It has been reported that the sweet-sensing gustatory neurons primarily receive regulatory inputs from neurons in the SEZ region (Marella et al., 2012 [↗](#); Song et al., 2023 [↗](#)). Therefore, we hypothesized that these neuropeptidergic neurons might function as satiety signals, imposing inhibitory effect on sweet sensation after adequate food ingestion.

To test this hypothesis, we selectively expressed the temperature-sensitive ion channel TrpA1 in these neurons and activated them at 30 °C, triggering the release of corresponding neuropeptides. Indeed, activation of LK-, TK-, hugin-, AstA-, CCHa1-, and CCHa2-expressing neurons robustly suppressed sweet taste sensitivity in both fed and starved flies (**Figure 1C-D** [↗](#); **also see Figure supplementary 2** [↗](#)). Moreover, LK-, TK-, and hugin-expressing neurons exerted particularly strong inhibition under satiated conditions (**Figure 1C-D** [↗](#); **also see Figure supplementary 2** [↗](#)). These results reveal a multilayered, state-dependent neuropeptidergic network that finely tunes gustatory processing and feeding behavior. In contrast, activating dopaminergic neurons enhanced sweet perception in sated flies as expected (**Figure 1C** [↗](#)). In starved flies, activating dopaminergic neurons elicited no change in PER likely due to a ceiling effect (**Figure 1D** [↗](#)).

In mammals and flies, circulating sugar levels are directly linked to satiety/hunger state (Lin et al., 2019 [↗](#); Qi et al., 2021 [↗](#); Yoon and Diano, 2021 [↗](#)). We then asked whether changes in circulating sugar levels exerted an effect on sweet sensation. Activating IPCs in the fly brain released *Drosophila* Insulin-like Peptides (DILPs), the fly analog of mammalian insulin (Brogiolo et al., 2001 [↗](#)), into the hemolymph and reduced circulating sugar levels (**Figure supplementary 3A** [↗](#)). As a result, such manipulation enhanced sweet sensitivity of fed flies (**Figure supplementary 3B-C** [↗](#)). Therefore, it was plausible that changes in circulating sugar might be the driver of changes in sweet sensation.

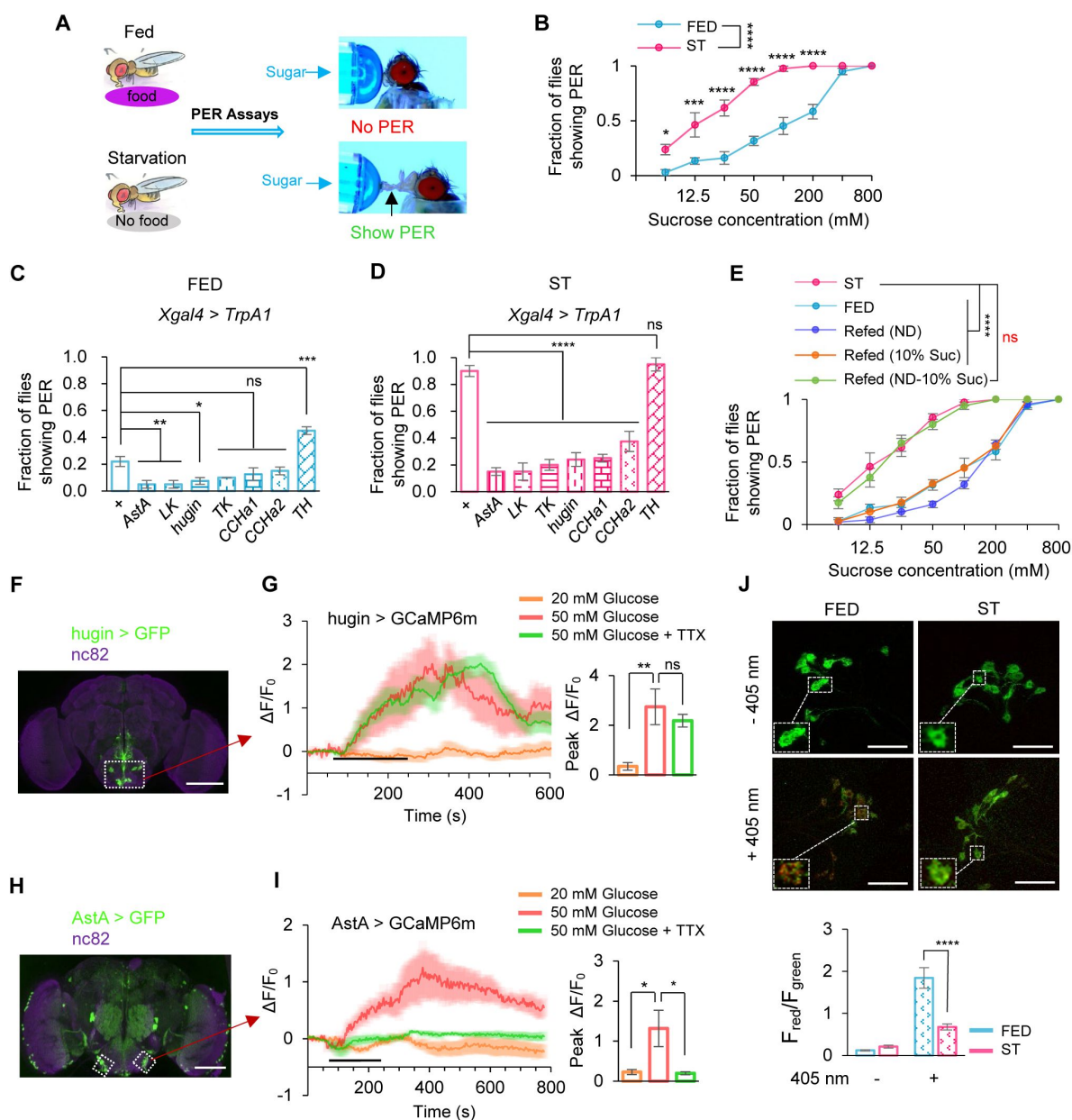


Figure 1.

***hugin*⁺ neurons were a novel glucose sensor in the fly brain.**

(A) Experimental procedure schematic. Newly eclosed flies were gathered and provided with regular food for 5 days, followed by a 24-hour period of starvation. Proboscis Extension Response (PER) assays were conducted. (B) Fraction of flies showing PER to different concentrations of sucrose ($n=4$ groups, each of 10 flies). (C-D) Fraction of flies of the indicated genotypes under 30 °C showing PER to 400 mM of sucrose ($n=4-5$ groups, each of 10 flies). (E) Fraction of indicated flies showing PER to different concentrations of sucrose (ND represent standard fly food) ($n=4-5$ groups, each of 10 flies). (F) *hugin* expression in the brain, illustrated by mCD8::GFP expression driven by *hugin*^{GAL4}. Scale bar represents 100 μ m. (G) Representative traces and quantification of ex vivo calcium responses of *hugin*⁺ neurons during the perfusion of glucose with or without TTX ($n=6-7$). Horizontal black bar represents the duration indicated glucose solution stimulation. (H) *AstA* expression in the brain, illustrated by mCD8::GFP expression driven by *AstA*^{GAL4}. Scale bar represents 100 μ m. (I) Representative traces and quantification of ex vivo calcium responses of *AstA*⁺ neurons during the perfusion of glucose with or without TTX ($n=6$). (J) Representative images of pre-photoconversion (pre-PC) and post-photoconversion (post-PC) CaMPARI signal in *hugin*-expressing neurons (upper). The Red:Green ratio represents intracellular Ca^{2+} concentrations (lower). Scale bar represents 100 μ m ($n=18-23$). ns, $P > 0.05$; * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$. Student's *t*-test, one-way and two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

Indeed, allowing starved flies to re-feed on normal fly diet or sucrose alone for 15 minutes both resulted in a significant decrease in sweet sensation (**Figure 1E**). To validate this effect across sweet modalities, we assessed responses to trehalose and fructose after sucrose refeeding and observed comparable reductions in sweet sensation (**Figure supplementary 4**). Conversely, when starved flies were supplied with fly diet deprived of sucrose (but with normal levels of polysaccharides, proteins, and lipids), there was no immediate change in sweet sensation after re-feeding (**Figure 1E**). These results demonstrate that sugar intake inhibits sweet sensation, probably via increasing circulating sugar levels.

Collectively, we thus hypothesized that the above neuropeptidergic neurons might directly sense an increase of circulating sugar levels and suppress sweet sensation in response. If so, these neurons should be activated by glucose, the main species of circulating sugars associated with feeding behavior in flies (Ugrankar et al., 2018), at the concentrations resembling sated state (>50 mM in fed flies (**Figure supplementary 5**)). We thus conducted calcium imaging experiments in fly brains in an *ex vivo* preparation (Qi et al., 2021). Among all the six groups of neuropeptidergic neurons we examined in the SEZ, only hugin⁺ neurons and AstA⁺ neurons could be activated by 50 mM glucose (**Figure 1F-I**, also see **Figure supplementary 6**). Remarkably, each of these populations subdivided into anatomically distinct SEZ subclusters (**Figure supplementary 7A**; **Figure supplementary 9A**), yet glucose-evoked Ca²⁺ responses occurred exclusively within a single subcluster (**Figure 1H,I**; **Figure supplementary 7B**; **Figure supplementary 9B**). This selective activation highlighted profound functional heterogeneity in nutrient sensing among these neuron groups. When direct synaptic transmission was blocked with TTX, hugin⁺ neurons, but not AstA⁺ neurons, still exhibited calcium responses towards 50 mM of glucose (**Figure 1G**, green), suggesting that hugin⁺ neurons are a direct glucose sensor. Furthermore, the level of circulating glucose dropped to ~ 20 mM in starved flies (**Figure supplementary 5**). We found that 20 mM of glucose did not activate hugin⁺ and AstA⁺ neurons (**Figure 1G and I**, orange), suggesting that these neurons can only be activated under sated conditions.

We also assessed the activity of hugin⁺ neurons *in vivo* using calcium-modulated photoactivatable ratiometric integrator (CaMPARI), a method employing a photoconvertible fluorescent protein that allows the imaging of integrated calcium activity (Fosque et al., 2015), under both sated and starved conditions. In the CaMPARI assay, when neurons are activated, leading to an increase in free calcium ions, exposure to 405-nm light induces a permanent green-to-red photoconversion (Fosque et al., 2015). We found an increased ratio of red/green fluorescence in hugin⁺ neurons in the sated state compared to that upon starvation (**Figure 1J**). These findings strongly imply that hugin⁺ neurons function as a direct satiety sensor, sensing the levels of circulating glucose and modulating sweet sensation in response.

Glucose activated hugin⁺ neurons via an ATP-sensitive potassium channel

We then investigated the cellular mechanism through which glucose could directly activate hugin⁺ neurons. In both mammals and flies, extracellular glucose is usually transported into the cytosol to activate glucose-sensitive neurons (Dus et al., 2015; Oh et al., 2019; Parton et al., 2007). When fly brains were pre-treated with phlorizin, a blocker of glucose transport (Dus et al., 2015; Oh et al., 2019), hugin⁺ neurons showed diminished calcium responses to glucose, suggesting activation of hugin⁺ neurons by glucose requires intracellular glucose (**Figure 2A**).

The transport of glucose mediated by glucose transporter 1 (Glut1) can activate certain neurons in the fruit fly brain (Dus et al., 2015; Oh et al., 2019). By specifically knocking down Glut1 expression in hugin⁺ neurons, we observed a significant reduction in calcium responses to glucose

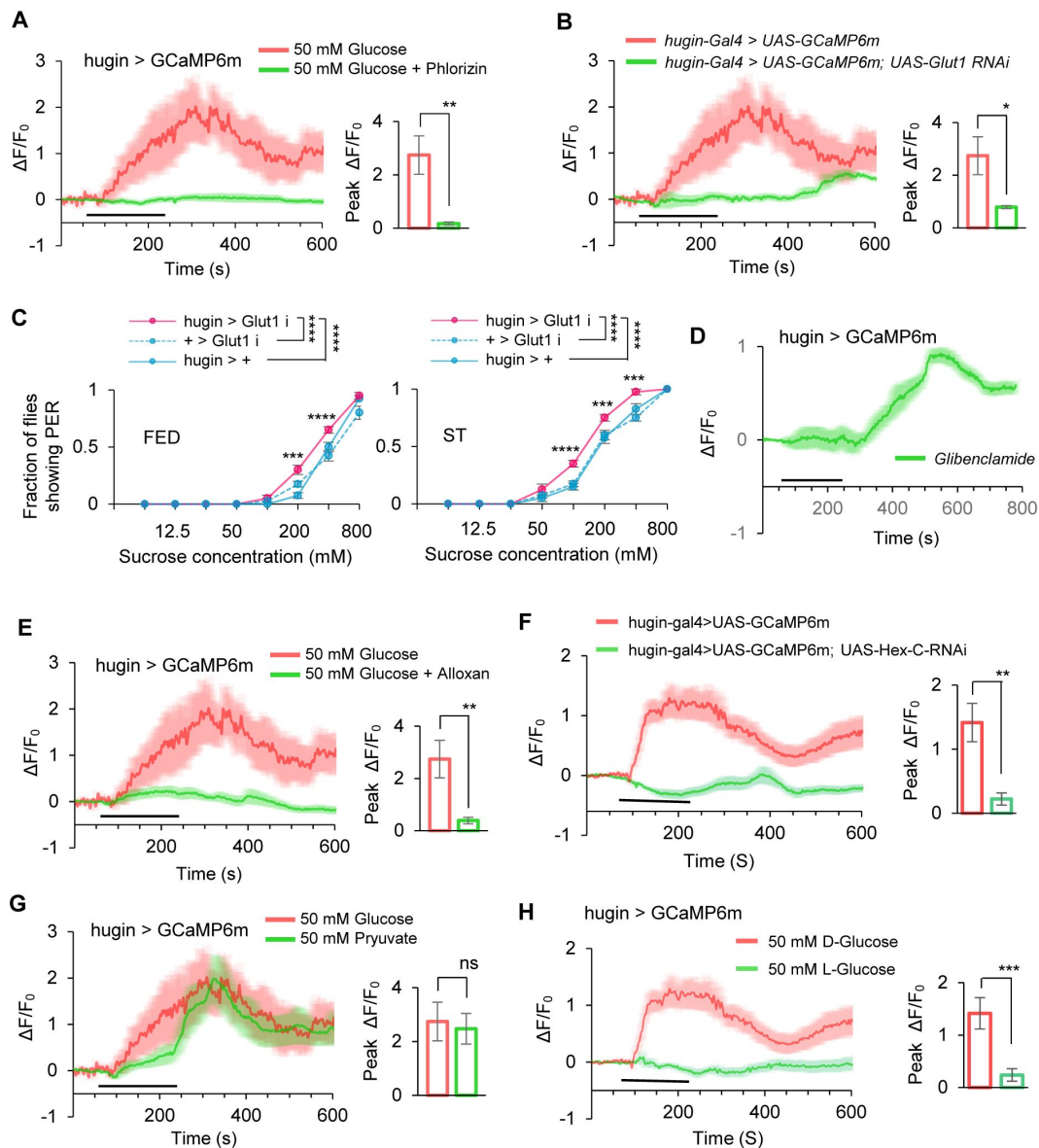


Figure 2.

Glucose activated *hugin*⁺ neurons via Glut1 and K_{ATP} channel.

(A) Representative traces and quantification of ex vivo calcium responses of *hugin*⁺ neurons during the perfusion of glucose with or without phlorizin (1 mM, n=6-7). Horizontal black bar represents the duration indicated glucose solution stimulation. (B) Representative traces and quantification of ex vivo calcium responses of *hugin*⁺ neurons in indicated flies during the perfusion of glucose. (C) Fraction of flies of the indicated genotypes showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). (D-E) Representative traces and quantification of ex vivo calcium responses of *hugin*⁺ neurons during the perfusion of glibenclamide (100 μ M, D), glucose with alloxan (10 μ M, E). (F) Representative traces and quantification of ex vivo calcium responses of *hugin*⁺ neurons in indicated flies during the perfusion of glucose. (G) Representative traces and quantification of ex vivo calcium responses of *hugin*⁺ neurons during the perfusion of pyruvate (50 mM, n=6). (H) Representative traces and quantification of ex vivo calcium responses of *hugin*⁺ neurons during the perfusion of D-glucose and L-glucose (50 mM, n=6) ns, P > 0.05; *P < 0.05; **P < 0.01; ****P < 0.0001. Student's t-test and two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

(**Figure 2B** [↗](#)). Consistently, these flies showed a slight yet significant increase in their PER responses to glucose (**Figure 2C** [↗](#)). Again, these results further confirm that glucose needs to be transported into hugin⁺ neurons to modulate neuronal activity and behavioral output.

In pancreatic β -cells and certain fly neurons (Oh et al., 2019 [↗](#); Parton et al., 2007 [↗](#); Wang et al., 2022 [↗](#)), intracellular glucose modulates cell excitability primarily through its effect on intracellular energy metabolism and ATP-sensitive potassium channels (K_{ATP}). We pre-treated fly brains with a K_{ATP} inhibitor (glibenclamide) and observed a significant calcium response in hugin⁺ neurons (**Figure 2D** [↗](#)). Hexokinase, a key enzyme in glucose metabolism that generates ATP, also plays a crucial role in this process. When hexokinase activity was restricted by alloxan, glucose-induced calcium response in hugin⁺ neurons was significantly suppressed (**Figure 2E** [↗](#)). Similarly, when hexokinase was specifically knocked down in hugin⁺ neurons, a comparable reduction in glucose-induced calcium responses was observed (**Figure 2F** [↗](#)). Pyruvate, the end product from glycolysis, can lead to ATP generation via the tricarboxylic acid cycle. We also found that pyruvate stimulation led to a significant calcium response in hugin⁺ neurons (**Figure 2G** [↗](#)). As a specificity control, nonmetabolizable L-glucose failed to activate hugin⁺ neurons (**Figure 2H** [↗](#)).

Collectively, these findings suggest that extracellular glucose directly enters hugin⁺ neurons through specific glucose transporter and subsequently modulate the activity of K_{ATP} through ATP production, ultimately leading to neuronal depolarization.

AstA⁺ neurons were direct downstream target of hugin⁺ neurons

Since hugin⁺ neurons were directly involved in glucose sensing and regulated sweet sensation (**Figure 1G** [↗](#) and **Figure supplementary 8A-B** [↗](#)), we sought to identify their downstream target. We then asked whether the hugin peptide had a direct effect on sweet sensation. Upon microinjection of synthetic hugin peptide, we observed a notable reduction in sweet sensation under both sated and starved states (**Figure 3A** [↗](#)). In line with this observation, following the injection of hugin peptide, we observed a significant reduction of sugar-induced calcium responses in sweet-sensing Gr5a⁺ neurons (**Figure 3B** [↗](#)). Thus, the inhibitory function of hugin⁺ neurons on sweet sensation was likely mediated by the secretion of hugin peptide.

There were two hugin receptors, PK2-R1 and PK2-R2, in fruit flies (Park et al., 2002 [↗](#)). We knocked down their expression by RNAi in the nervous system and then tested flies' PER to sugar. Knocking down PK2-R1 but not PK2-R2 significantly enhanced sweet perception in both sated and starved states (**Figure 3C-D** [↗](#)), suggesting a crucial role of PK2-R1 in receiving the input of hugin⁺ neurons in satiety sensing. To validate this finding, we further tested PK2-R1 null mutants and observed a similar enhancement in sweet sensitivity (**Figure supplementary 10** [↗](#)). Interestingly, we did not observe any expression of PK2-R1 in the proboscis where gustatory sensory neurons were distributed, indicating that sweet-sensing Gr5a⁺ neurons could not directly receive input from hugin⁺ neurons via PK2-R1 (data not shown).

AstA⁺ neurons served as a plausible relay between hugin⁺ neurons and sweet-sensing Gr5a⁺ neurons, especially based on a clear co-expression of hugin receptor PK2-R1 and AstA in the SEZ region of the fly brain (**Figure 3E** [↗](#)). Notably, these double-labeled neurons corresponded precisely to the AstA subcluster that exhibited glucose-specific Ca^{2+} responses (**Figure 1H** [↗](#)). We also used optogenetic tools to further confirm that hugin⁺ neurons acted upstream of AstA⁺ neurons. We expressed the light-sensitive neuronal activator CsChrimson in hugin⁺ neurons and calcium indicator GCaMP6m in AstA⁺ neurons, respectively (Klapoetke et al., 2014 [↗](#)). As shown in **Figure 3F** [↗](#), opto-activation of hugin⁺ neurons led to a robust calcium transient in AstA⁺ neurons.

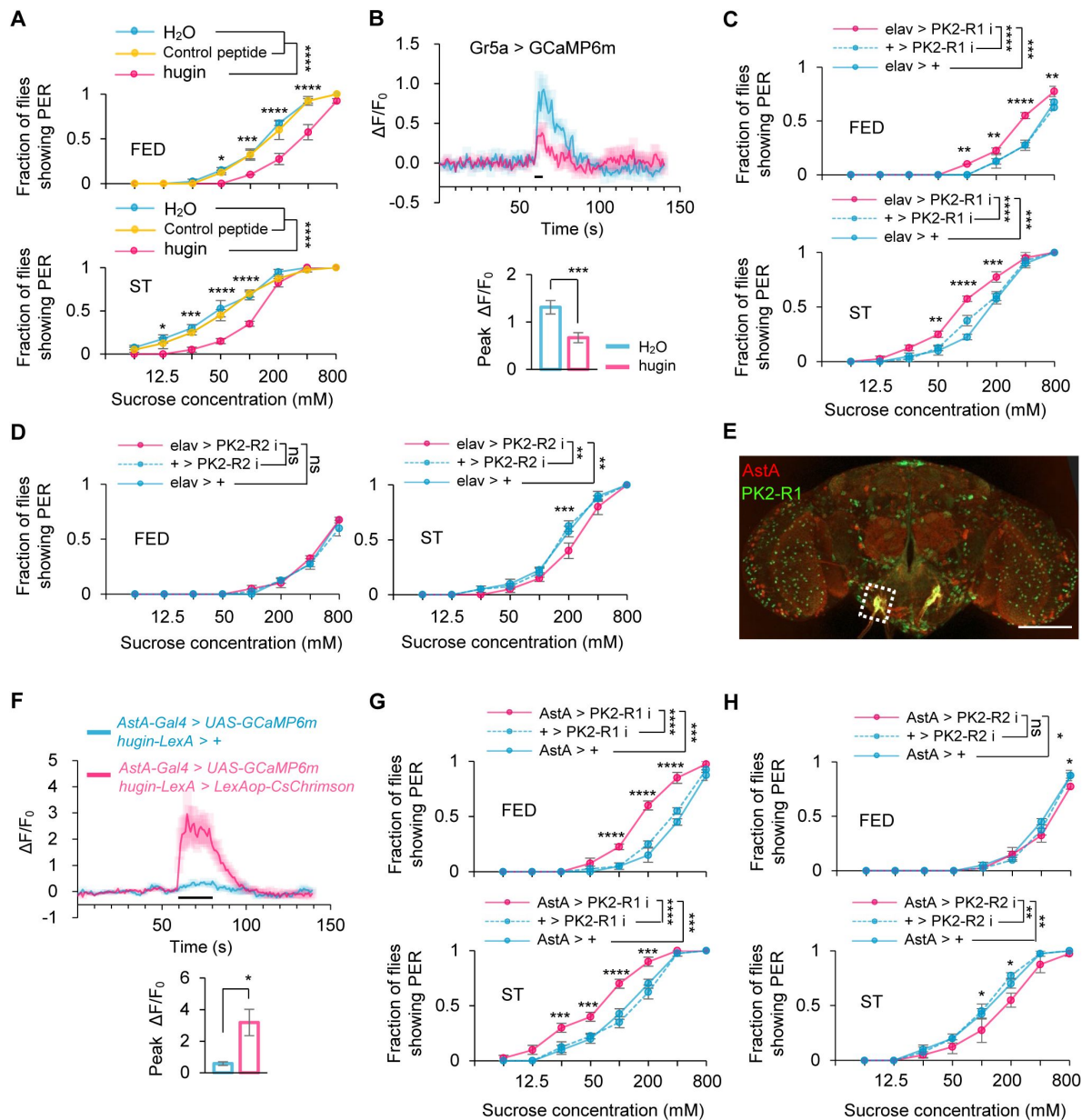


Figure 3.

hugin⁺ neurons activated AstA⁺ neurons through PK2-R1.

(A) Fraction of flies showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). Flies were injected with saline or synthetic hugin for 30 minutes before the assay. (B) Representative traces (upper) and quantification (lower) of peak calcium transients of Gr5a⁺ neurons in indicated flies upon 5% sucrose after injection of synthetic hugin (n=6). Horizontal black bar represents the duration sucrose stimulation. (C-D) Fraction of flies of the indicated genotypes showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). (E) Co-localization (dashed box) of PK2-R1⁺ neurons (green) and AstA⁺ neurons (red) in the SEZ region. Scale bar represents 100 μ m. (F) Representative traces (upper) and quantification (lower) of peak calcium transients of AstA⁺ neurons after the photo-activation of hugin⁺ neurons (n=6) from in vivo calcium imaging. Horizontal black bar represents the duration of red-light stimulation. (G-H) Fraction of flies of the indicated genotypes showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). ns, P > 0.05; *P < 0.05; **P < 0.01; ***P < 0.001; ****P < 0.0001. Student's *t*-test and two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

We then investigated whether the hugin-AstA circuitry affected sweet sensation. As expected, knocking-down PK2-R1 (but not PK2-R2) in AstA⁺ neurons led to a significant increase in sugar-induced PER (**Figure 3G-H**). Therefore, AstA⁺ neurons are the direct downstream target of hugin⁺ neurons in satiety sensing. hugin⁺ neurons can directly sense elevated circulating glucose levels, promote the release of neuropeptide hugin, and then activate AstA⁺ neurons via its cognate receptor PK2-R1.

hugin⁺ neurons are distributed in both the brain and the ventral nerve cord (VNC). The hugin⁺ neurons in the VNC sense sugar and inhibit feeding behavior by suppressing the activity of DH44 neurons (Oh et al., 2021). To determine which hugin⁺ neurons regulate the activity of AstA neurons and sweet sensation, we performed the following experiments: 1, Physically severing the neural connection between the brain and the VNC: After severing the neural connection, activation of hugin⁺ neurons were still able to downregulate the PER behavior, suggesting that hugin neurons in the SEZ regulated sweet perception (**Figure supplementary 11A-B**). 2, Optogenetic experiments in isolated brains: We dissected the fly brains in **Figure 3F**, and upon light stimulation, we observed a significant calcium response in AstA neurons (**Figure supplementary 11C**). This indicates that hugin⁺ neurons located in the SEZ but not VNC regulate the activity of AstA neurons. Therefore, we can infer that hugin neurons in the SEZ are responsible for sugar sensing, energy signal transmission, and the output of PER behavior.

AstA⁺ neurons directly inhibited Gr5a⁺ neurons

Activation of AstA⁺ neurons suppressed sweet sensation, resembling the effect of activating hugin⁺ neurons (Hergarden et al., 2012)(**Figure supplementary 8C-D**). We also evaluated the behavioral effect of AstA peptide. Similar to hugin, the microinjection of AstA resulted in a reduction in sweet sensitivity in fruit flies (**Figure 4A**) and the activity of Gr5a⁺ neurons (**Figure 4B**). We then speculated whether AstA⁺ neurons and AstA peptide might function directly on Gr5a⁺ neurons. The cell bodies of sweet-sensing Gr5a⁺ neurons are located in the proboscis, whereas their axons extend to the SEZ region of the fly brain (Dahanukar et al., 2007). We observed distributions of the projections of AstA⁺ neurons and Gr5a⁺ neurons both in the SEZ region (**Figure 4C**), suggesting that Gr5a⁺ neurons might directly receive AstA signal in this region.

AstA has two cognate receptors in fruit flies, AstA-R1 and AstA-R2 (Larsen et al., 2001). Co-labeling experiments in the proboscis confirmed the expression of AstA-R1 in Gr5a⁺ neurons (**Figure 4D**, **arrows**), while AstA-R2 was not expressed in the proboscis (data not shown). Functionally, knockdown of AstA-R1 specifically in Gr5a⁺ neurons, but not that of AstA-R2, significantly enhanced sweet sensation of flies (**Figure 4E-F**). Similar results were observed in AstA-R1 knockout flies (**Figure supplementary 12**).

Using optogenetic tools, we also confirmed the function of hugin⁺ neurons and AstA⁺ neurons on Gr5a⁺ neurons. Activating hugin⁺ neurons and AstA⁺ neurons both significantly reduced sugar-induced responses of Gr5a⁺ neurons (**Figure 4G-H**). Collectively, these results suggest that satiety-sensing hugin-AstA circuitry can directly modulate sweet sensation of flies via suppressing the activity of Gr5a⁺ gustatory neurons.

hugin-AstA circuitry suppressed food consumption

Collectively, the hugin-AstA circuitry senses satiety signals and inhibits sweet sensation. Considering the crucial role of sweet sensation in regulating feeding behavior, we hypothesized that hugin-AstA circuitry was also involved in the regulation of food consumption.

We expressed a temperature-sensitive blocker of synaptic transmission (Shibire^{ts1}) in hugin⁺ neurons or AstA⁺ neurons (Kitamoto, 2001) and assayed the feeding behavior of flies under both permissive temperature (**Figure supplementary 13A**, **blue**) and non-permissive temperature

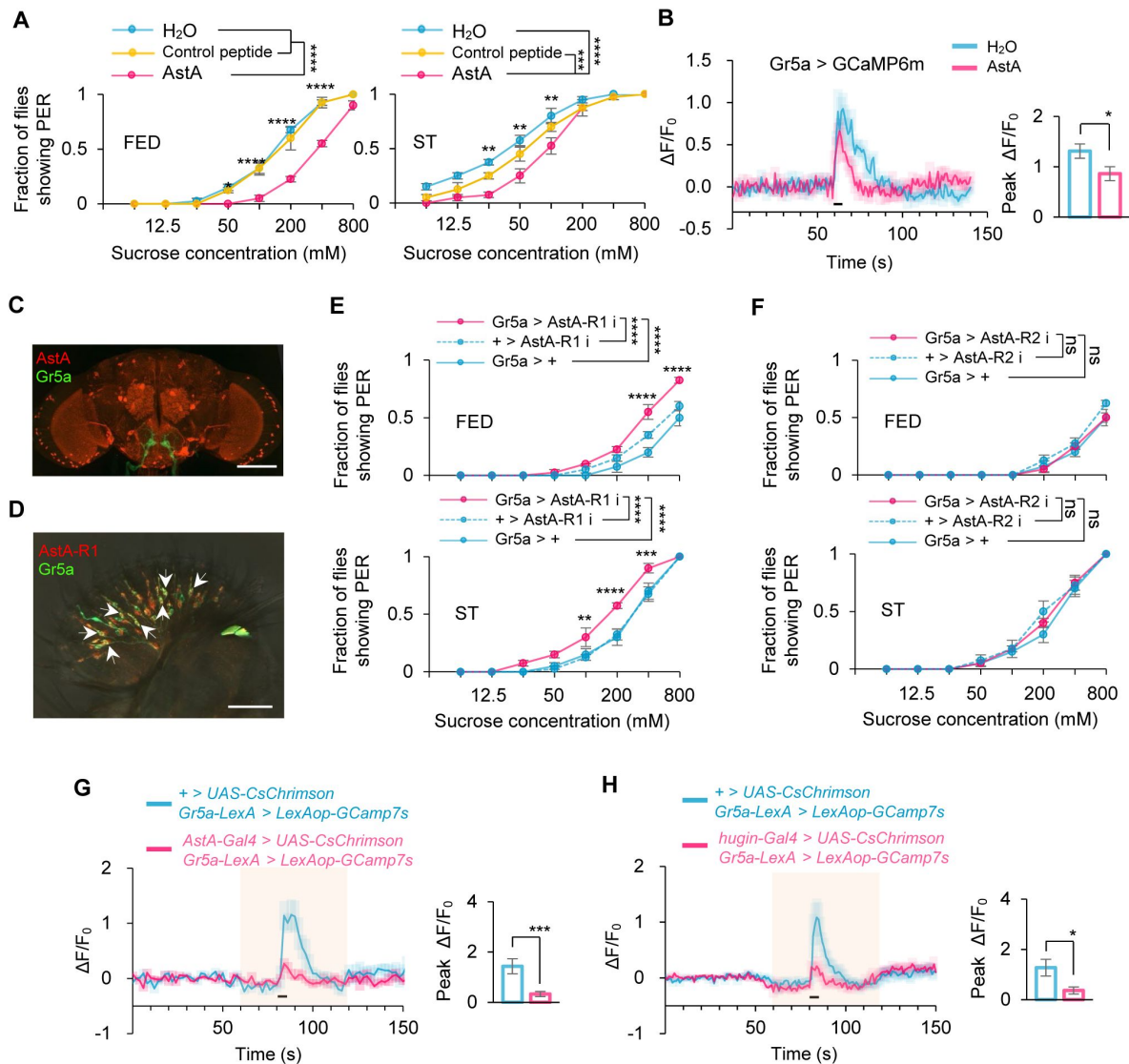


Figure 4.

AstA⁺ neurons inhibited Gr5a⁺ neurons through AstA-R1.

(A) Fraction of flies showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). Flies were injected with saline or synthetic AstA for 30 minutes before the assay. (B) Representative traces (left) and quantification (right) of peak calcium transients of Gr5a⁺ neurons in indicated flies upon 5% sucrose after injection of synthetic AstA (n=6). Horizontal black bar represents the duration sucrose stimulation. (C) Localization of Gr5a⁺ neurons (green) and AstA⁺ neurons (red) in the brain. Scale bar represents 100 μ m. (D) Co-localization of Gr5a⁺ neurons (green) and AstA-R1⁺ neurons (red). Arrows point to the neurons co-expressing the two receptors. Scale bar represents 50 μ m. (E-F) Fraction of flies of the indicated genotypes showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). (G-H) Representative traces (left) and quantification (right) of peak calcium transients of Gr5a⁺ neurons in indicated flies upon 5% sucrose during the process of photo-activation of AstA⁺ neurons (G, n=6) and hugin⁺ neurons (H, n=7-8). Shadows represents the duration of red-light stimulation. Horizontal black bar represents the duration of sucrose stimulation. ns, P > 0.05; *P < 0.05; ****P < 0.001; ****P < 0.0001. Student's *t*-test and two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

(**Figure supplementary 13A** [↗](#), *red*). We found that inhibiting the activity of either hugin⁺ neurons or AstA⁺ neurons significantly increased food consumption (**Figure supplementary 13B-C** [↗](#)), demonstrating the inhibitory role of the hugin-AstA circuitry on feeding behavior.

We also examined the role of hugin and AstA receptors in feeding behavior by testing *PK2-R1* and *AstA-R1* gene knockout flies. The results showed that knockout of both receptors enhanced feeding behavior (**Figure supplementary 13D** [↗](#)), consistent with increased sweet perception upon genetic manipulation of these two receptors (**Figure 3** [↗](#)-[↗](#)). Moreover, these receptor knockout flies exhibited increased lipid storage, which was likely a consequence of their enhanced feeding behavior (**Figure supplementary 13E** [↗](#)). Additionally, we knocked down the expression of AstA-R1 in sweet sensing Gr5a⁺ neurons, and found that these flies exhibited a significant increase in food consumption (**Figure supplementary 13F** [↗](#)), also likely due to the enhanced sweet sensation (**Figure 4E** [↗](#)).

Mammalian homolog of hugin was also a putative satiety sensor

Our present work identified hugin-AstA circuitry as a novel satiety sensor and feeding suppressor in fruit flies. Fruit flies and mammals share highly conserved regulatory elements on feeding behavior and energy homeostasis (Song et al., 2023 [↗](#); Stocker, 2004 [↗](#); Yarmolinsky et al., 2009 [↗](#)). We thus asked whether this novel satiety-sensing mechanism was also conserved in the mammalian system,

In rodent models, neuromedin U (NMU), a analogue of hugin, was also found to be an appetite suppressor (Melcher et al., 2006 [↗](#); Schlegel et al., 2016 [↗](#)). We found that synthetic NMU could suppress sweet perception when injected into flies' thorax (**Figure supplementary 14** [↗](#)), highlighting the analogy of fly hugin and mammalian NMU. In mouse models, we observed a significant increase in NMU levels in the blood following glucose feeding (**Figure 5A** [↗](#)). Using the sucrose preference test (SPT) to measure sweet preference, we noted that NMU injection reduced sugar preference in mice, phenocopying the effect of glucose feeding (**Figure 5B-C** [↗](#)). Conversely, NMU knockout enhanced sweet sensation in mice (**Figure 5D** [↗](#)). These data demonstrated that both NMU gene and protein plays a crucial role in regulation of sweet sensation.

Neuropeptide NMU is secreted by a group of neurons located in the VMH region of hypothalamus (Melcher et al., 2006 [↗](#)). To investigate whether NMU⁺ neurons functioned as an energy sensor in the mouse brain, we injected AAV-EF1α-DIO-GCaMP6m-WPRE into the VMH of NMU-Cre mice, specifically expressing GCaMP6m in NMU⁺ neurons. We then monitored the dynamic activity of VMH^{NMU} neurons during glucose stimulation by using fiber photometry (**Figure 5E** [↗](#)). As expected, these neurons showed significant calcium responses following glucose administration in mice (**Figure 5F-G** [↗](#)). To confirm the specificity of GCaMP6m expression in NMU⁺ neurons, we co-expressed a Cre-dependent mCherry reporter in the VMH of NMU-Cre mice and performed fluorescence in situ hybridization for NMU mRNA. Colocalization analysis demonstrated that GCaMP6m fluorescence was restricted exclusively to NMU⁺ cells (**Figure supplementary 15A** [↗](#)).

Next, we conducted calcium imaging of brain slices *in vitro* to further assess these neurons' ability to sense glucose (**Figure 5H** [↗](#)). Similar to hugin neurons in the fly brain, VMH^{NMU} neurons in the mouse brain exhibited strong calcium responses to glucose stimulation, which could not be inhibited by TTX (**Figure 5I** [↗](#)), indicating their direct glucose-sensing capability. Also similar to flies, glucose activated these mouse neurons via entering the cells and modulating K_{ATP} channels, as evidenced by the inhibition of calcium responses by alloxan and phlorizin (**Figure 5J** [↗](#)).

Calbindin 2-positive neurons (Calb2) in the rostral nucleus of the solitary tract (rNST) selectively respond to sweet tastants, facilitating sweet signal from the periphery into the brain (Jin et al., 2021 [↗](#)). Because attenuated sweet sensation should correspondingly diminish central sweet-signal transmission, we asked whether NMU modulates this relay. To assess potential anatomical connectivity between VMH NMU⁺ neurons and rNST Calb2⁺ neurons, we employed a dual-virus

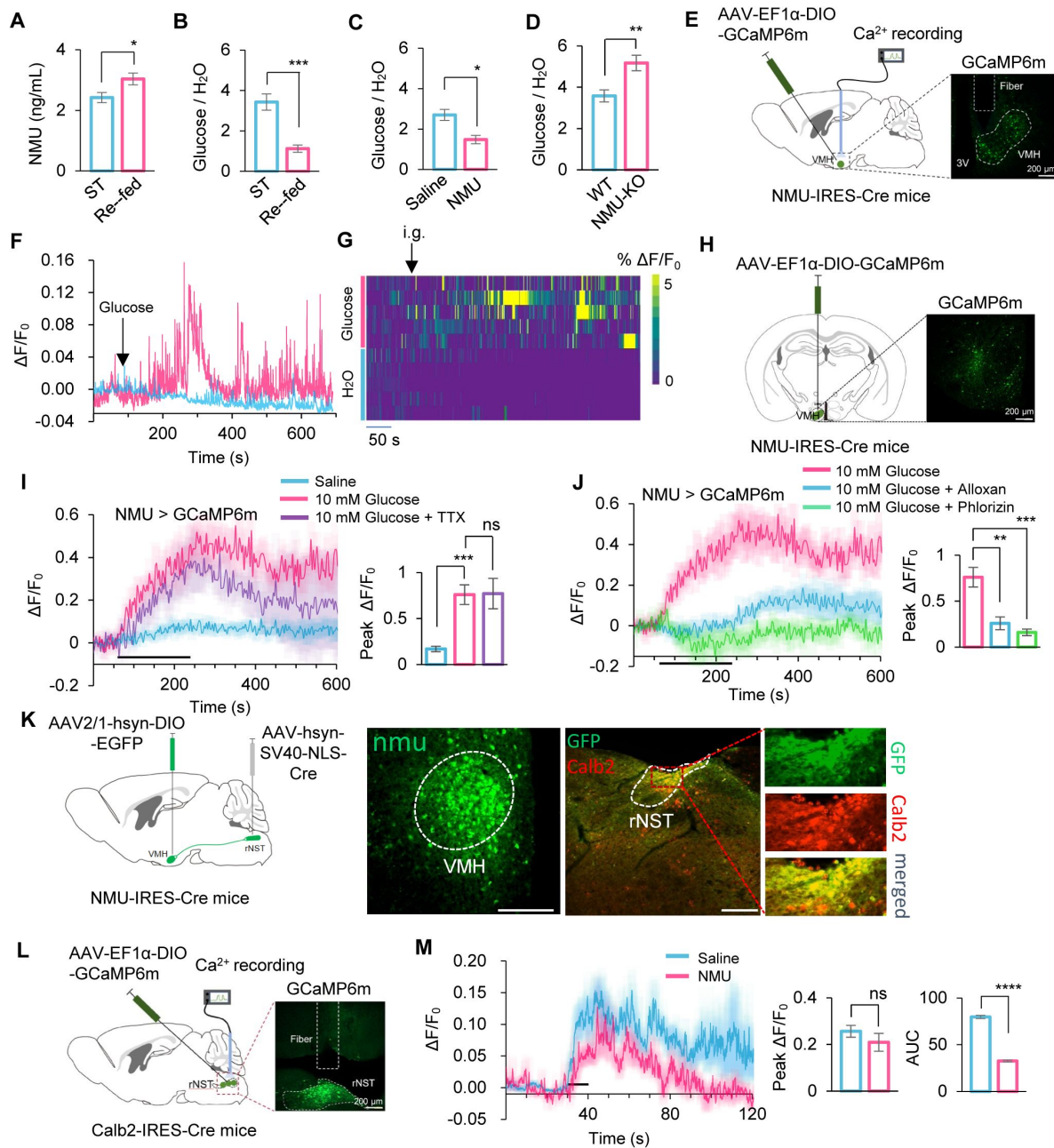


Figure 5.

NMU⁺ neurons were the central energy sensor in mouse.

(A) Blood NMU levels of starved mice re-fed with 20% sucrose (n=6). (B) Two-bottle preference tests for starved mice re-fed with sucrose (n=7). (C) Two-bottle preference tests for starved mice with or without intraperitoneal injection of NMU peptide (YFLFRPRN-NH₂, 4.5μm/kg) (n=7). (D) Two-bottle preference tests for indicated starved mice. (E) Fiber photometry to record the calcium dynamics of NMU⁺ neurons in the VMH with GCaMP6m. (F-G) Representative trace (left) and heatmaps (right, n=5) showing calcium dynamics of NMU⁺ neurons in the VMH in response to glucose perfusion. (H) Experimental approach to assess calcium signaling in NMU⁺ neurons in the VMH in vitro. (I-J) Representative traces and quantification of ex vivo calcium responses of NMU⁺ neurons during the perfusion of glucose with or without TTX (I), Alloxan (J) and Phlorizin (J) (n=6-7). Horizontal black bar represents the duration indicated glucose solution stimulation. (K) Anterograde tracing the downstream target of NMU neurons. (L) Fiber photometry to record the calcium dynamics of Calb2⁺ neurons in the rNST with GCaMP6m. (M) Representative traces and quantification of calcium responses of Calb2⁺ neurons during glucose feeding with or without NMU injection. ns, P > 0.05; *P < 0.05; **P < 0.01; ***P < 0.001; ****P < 0.0001. Student's *t*-test and one-way ANOVA followed by post hoc test with Bonferroni correction was used for multiple comparisons when applicable.

tracing strategy. In NMU-Cre mice, we injected AAV9-EF1 α -DIO-EGFP into the VMH, which revealed dense GFP⁺ projections in close apposition to Calb2⁺ neurons in the rNST (**Figure supplementary 15B** [↗](#)). To further confirm this projection is monosynaptic, we combined anterograde tracing by co-injecting AAV2/1-EF1 α -DIO-EGFP into the VMH and AAV2/1-hsyn-SV40-NLS-Cre into the rNST. This approach yielded GFP expression in rNST Calb2⁺ neurons, supporting a direct VMH → rNST NMU⁺ → Calb2⁺ synaptic link (**Figure 5K** [↗](#)).

To evaluate the functional consequence of NMU on Calb2⁺ neuronal activity during sweet stimulation, we delivered AAVs encoding Cre-dependent GCaMP6m into the rNST of Calb2-Cre mice, thereby enabling real-time calcium imaging in Calb2⁺ neurons (**Figure supplementary 15C** [↗](#)). Using fiber photometry, we recorded calcium responses to sugar stimuli, both with and without NMU injection (**Figure 5L** [↗](#)). We found that NMU injection significantly suppressed the calcium responses to sugar, aligning with observed behavioral phenotypes (**Figure 5M** [↗](#)). These data indicated that the neuropeptide NMU plays a role in regulating sweet taste perception during sugar feeding.

These findings indicate that NMU⁺ neurons can indeed be a putative satiety sensor in the mouse brain that suppresses sweet perception, shedding light on understanding the conserved neural mechanism of feeding regulation in mammals including human.

Discussion

In summary, we identified hugin-AstA circuitry as a novel satiety sensor in the fly brain. It directly detects circulating sugar levels, inhibits sweet taste perception, and terminates feeding, helping to maintain the homeostasis of internal energy state (**Figure supplementary 16** [↗](#)). Previous research has shown that hugin⁺ neurons project to the pharyngeal muscles and the ring gland, playing a role in regulating both feeding and sleep, and these neurons are also part of a neural circuit that modulates taste-mediated feeding behavior. (Melcher and Pankratz, 2005 [↗](#); Miroshnikow et al., 2020 [↗](#); Oh et al., 2021 [↗](#)). AstA⁺ neurons convey carbohydrate-satiety signals and shift dietary preference from sugar to protein (Hentze et al., 2015 [↗](#); Hergarden et al., 2012 [↗](#)). However, it was unclear whether they could sense the body's energy state, how they interacted with each other, or how they influenced sensory/motor neurons to regulate feeding. Our research has clarified the relationship between hugin and AstA: hugin functions upstream, sensing glucose, while AstA acts downstream, directly modulating sweet-sensing Gr5a⁺ neurons to regulate sugar perception and feeding behavior. This discovery establishes a mechanistic link between glucose sensing and sweet sensation regulation, enhancing our understanding of how energy states influence nutrient sensing and feeding behavior.

Recent studies have highlighted the complex interplay of neural circuits and neuropeptides that regulate satiety and feeding behavior in *Drosophila*. For instance, dopaminergic and neuropeptide F (NPF) circuits play crucial roles in modulating the activity of Gr5a⁺ sweet-sensing neurons during hunger and refeeding. These studies showed that dopamine signaling adjusts the temporal integration of PER-driving inputs under fed conditions, while NPF enhances Gr5a sensitivity during starvation (Inagaki et al., 2012 [↗](#); Inagaki et al., 2014 [↗](#); Marella et al., 2012 [↗](#)). Our findings complement these established mechanisms by identifying the hugin-AstA pathway as a central satiety sensor that directly detects circulating glucose levels and inhibits sweet taste perception to prevent overconsumption. Moreover, the hugin-AstA circuitry fits within a broader regulatory framework of satiety signaling in *Drosophila*. Peripheral satiety signals, such as enteroendocrine NPF⁺ cells, detect luminal sugar and suppress feeding via hormonal release (Malita et al., 2022 [↗](#)). Central hunger circuits, including dopaminergic neurons, adjust Gr5a responsiveness according to the energy state (Inagaki et al., 2012 [↗](#)). Our study reveals that the hugin-AstA loop acts as a

central brake on Gr5a activity following sugar intake, thus closing a feedback loop that protects against excessive sugar ingestion. Together, these multilayered mechanisms—from gut to brain—ensure robust regulation of sweet perception and energy balance.

Although multiple regulatory elements of energy homeostasis have been identified in both flies and mammals, our present work reveals a novel mechanism that links energy sensing to taste modulation. The energy sensor comprises three types of neurons: hugin⁺ neurons as the sensor, AstA⁺ neurons as a relay, and sweet-sensing Gr5a⁺ neurons as the behavioral modulator. Given the highly conserved regulatory mechanism of energy homeostasis across animal species (**Figure 5A-D**), our work paves a way to fully understand how internal state sensing and behavioral alternations are tightly coupled in more complex organisms, and how they are disrupted in metabolic disorders.

In addition to the novel mechanism we have uncovered in *Drosophila*, it is interesting to consider how conserved these pathways might be in more complex organisms like mammals, particularly concerning the function of the mammalian homologues, NMU (Melcher et al., 2006). NMU is known to play critical roles in regulating energy homeostasis, feeding behavior, insulin secretion, and immune responses in mammals (Teranishi and Hanada, 2021). Our study has shown that NMU neurons may serve as an energy sensor in mammals as well, which regulated nutrient sensing, much like the hugin neurons in *Drosophila*. Given the functional parallels between the hugin-AstA circuitry in flies and NMU signaling in mammals, it is plausible that NMU may interfere with taste signaling to the brain and modulate the brain's reward circuits, affecting other behaviors beyond feeding.

Additionally, our findings open up the possibility of using NMU or its pathways as therapeutic targets for appetite suppression and obesity treatment. Recent developments in appetite-regulating peptides, such as glucagon-like peptide 1 (GLP-1) receptor agonists, have shown efficacy in reducing food intake and promoting weight loss in humans (Aldawsari et al., 2023). Given NMU's role in suppressing sweet taste perception and energy regulation, the NMU system could be a complementary or alternative target for anti-obesity interventions. Exploring how modulating NMU activity could synergize with GLP-1 therapies may provide new strategies for controlling appetite and managing energy balance.

In conclusion, our discovery of the hugin-AstA circuitry in flies provides a foundation for understanding the broader principles of energy sensing and behavioral regulation across species. Further studies in mammals will be essential to explore the potential roles of NMU neurons in these processes and their therapeutic relevance in metabolic disorders.

Materials and methods

Animals

Fly

Flies were maintained on a standard fly diet under conditions of 60% humidity and 25 °C with a 12-hour light and 12-hour dark cycle. Virgin female flies were chosen immediately after emerging and housed in standard fly food (ND) vials, with 20 flies per vial, for a period of 5-6 days before experiments. For temperature-sensitive manipulations (dTrpA1, Shibire^{ts}), flies were reared at 18 °C for 7–8 days, then transferred to 30 °C for 30 min to activate or inhibit targeted neurons; behavioral assays (PER or feeding) were conducted continuously at 30 °C. For starvation, the flies were kept on 2 % Agar for 12-hour.

The following UAS-RNAi lines: *UAS-AstA-R1 RNAi* (#27280), *UAS-AstA-R2 RNAi* (#25935), *UAS-PK2-R1 RNAi* (#29624), *UAS-PK2-R2 RNAi* (#28781), *UAS-CD8-GFP*, *LexAop-CD2-RFP* (#67093), *UAS-GCaMP6m* (#42748), *UAS-CaMPARI* (#58764), *Gr5a-GAL4* (#57592), *Gr5a-LexA* (#93014), *ilp2-GAL4* (#37516), *elav-GAL4* (#25750) were obtained from the Bloomington Drosophila Stock Center at Indiana University. *TH-KO*, and all the other Gal4 and LexA lines were from Yi Rao (Deng et al., 2019) (Capital Medical University, Beijing). *UAS-dTrpA1* and *UAS-Shibire^{ts1}* were from David Anderson (Caltech). *UAS-GCaMP6m*; *LexAop-CsChrimson* was from Yufeng Pan (Southeast University, Nanjing). *LexAop-Chrimson* was from Wei Zhang (Tsinghua University, Beijing).

Mouse

Male C57BL/6J mice at the age of four weeks were procured from cyagen. The mice were kept in standard laboratory conditions at a temperature of 25 °C with a 12:12 light/dark cycle. All experimental procedures were conducted in compliance with the guidelines set by the Laboratory Animal Welfare and Ethics Committee of Chongqing University (CQU-IACUC-RE-202401-004), adhering to both national and international standards.

Behavioral assays (Fly)

PER was evaluated using a method outlined in a previous study. Briefly, individual flies were gently aspirated and positioned in a 200 µL pipette tip, allowing only the head and proboscis to protrude while the dorsal thorax was lightly affixed to prevent escape. After a 3-minute acclimation period, each fly was first presented with 0.5 µL water droplets on the labellum twice. Flies that responded by extending their proboscis were allowed to drink and were subsequently excluded from the experiment. Flies that did not extend their proboscis in response to water were then stimulated sequentially with ascending sucrose concentrations (6.25–800 mM). Each concentration was delivered as two brief (<1 s) touches of a 0.5 µL droplet to the labellum, immediately withdrawn to prevent ingestion. A full proboscis extension, defined as the complete uncoiling of the rostrum beyond the labellum plane, was scored as positive. Partial or delayed (>1 s) extensions were not considered positive responses. Responsiveness at each concentration was recorded if at least one full extension occurred in the two trials. For **Figure supplementary 11**, the connection between the brain and VNC were cut off by a dissecting scissors before the PER assay.

Feeding was assessed as previously described (Sun et al., 2017). Briefly, 5-day-old flies underwent a 12-hour starvation period on 2% agar and were then moved to a new vial with test food containing 0.5% Brilliant Blue (MACKLIN, China) for 10 minutes. After a rapid freeze at −20 °C, flies were decapitated in PBS, homogenized, and centrifuged (13,000 g * 5 min). The resulting supernatants were diluted with PBS to a total volume of 1 ml, and absorbance was measured at 620 nm.

Sucrose Preference Test

The two-bottle preference (TBP) behavioral test was conducted in a standard mouse cage. Two bottles, one containing water and the other with a 10 mM sucrose solution, were placed on top of the cage. Each tested mouse was individually caged and adopted to two bottles containing distilled water or a 10 mM sucrose solution for one week. Throughout the training period, the positions of the bottles were alternated every 24 hours to mitigate potential position-related biases. Prior to the TBP tests for sucrose, mice underwent a 12-hour pre-starvation period, followed by re-fed sucrose (20%) for 1 hour or intraperitoneal (IP) administration of NMU (YFLFRPRN-NH₂, 4.5 µm/kg). The consumption of water and sucrose solution over a 2-hour period was meticulously recorded. The sweetener preference ratio was then calculated by dividing the weight of sweetener consumed by the weight of water consumed.

NMU measurement

Blood was obtained from the lower jaw of the mice and clotted at room temperature for 1 hour. The samples were then centrifuged (1,500g, 4 °C) for 10 minutes and the supernatants were collected for NMU measurement. All the manipulations were according to the manufacturer's instructions (BMASSY,74030).

Hemolymph extraction and glucose measurement

Groups of 40 ice-anesthetized flies were decapitated and placed into a perforated 0.5 ml tube, which was nested inside a 1.5 ml collection tube. Samples were centrifuged at $2,500 \times g$ for 10 min at 4 °C, yielding approximately 2 μ l of hemolymph per batch. Hemolymph was pooled to 4 μ l per assay (n = 8 independent samples per group) and immediately analyzed for glucose concentration using the Solarbio BC2500 kit (Solarbio, China) according to the manufacturer's instructions.

Microinjection

Flies were delicately positioned and secured within a 200 μ l pipette tip, ensuring their heads were directed toward the tip's end. Subsequently, the tip was opened by cutting, exposing the heads and a section of the thorax. About 20 nl of sterile saline, either with or without synthesized peptides (hugin: SVPFKPRL-NH₂, 1 mg/ml (1.1 mM); AstA: LPVYNFGL-NH₂, 1mg/ml (1.1 mM); NMU: YFLFRPRN-NH₂, 1mg/ml (0.9 mM); Control peptide: DYKDDDDKYPYDVPDYA, 1mg/ml (0.48 mM), were injected into the thorax of these flies using a glass micropipette and a microinjector (3-000-207, Drummond Scientific Company Instruments). The glass micropipette was created from thick-walled borosilicate capillaries (3-000-203-G/X, Drummond Scientific Company Instruments).

Immunofluorescence staining

Fly brains were carefully dissected in PBS on ice and then fixed in 4% formaldehyde for 60 minutes. Following fixation, the brains underwent permeabilization and blocking using Dilution/Blocking Buffer (PBS containing 10% Calf Serum and 0.5% Triton X-100) for 2 hours at room temperature. Subsequently, the samples were immersed in the appropriate primary antibodies in Dilution/Blocking Buffer for 24 hours at 4 °C. Afterward, the samples were subjected to a 60-minute wash with Washing Buffer (0.5% Triton X-100 in PBS) four times at room temperature and were then incubated with secondary antibodies for 24 hours at 4 °C. The samples underwent three additional washes with Washing Buffer before being mounted in Fluoroshield (Sigma-Aldrich).

Images were acquired using a scanning confocal microscope with Olympus objectives (20 $\times$ /0.7 and 40 $\times$ /0.95w). Antibodies were employed at the following dilutions: mouse anti-nc82 (1:100, DSHB), rabbit anti-GFP (1:500, Abcam), Alexa Fluor 647 goat anti-mouse (1:500, Invitrogen), and Alexa Fluor 488 goat anti-rabbit (1:500, Invitrogen).

In situ hybridization

Mice were anesthetized and perfused with 4% paraformaldehyde (PFA) dissolved in phosphate-buffered saline (PBS, pH 7.4). Brains were carefully removed from the skull and post-fixed in 4% PFA at 4 °C overnight. Subsequently, the brains were transferred to a 20% sucrose solution (in 1 $\times$ PBS) at 4 °C until they were ready for sectioning. Twenty-micrometer-thick sections were obtained using a cryostat. The detection of nmU was performed using a fluorescent in situ hybridization technique (RNAscope, Pinpointase) according to the manufacturer's instructions. Sections were mounted on SuperFrost Plus Gold slides (ThermoFisher) and briefly rinsed in autoclaved Millipore water. They were then subjected to gradient dehydration in 50%, 75%, and 100% ethanol for 5 minutes each. A hydrophobic barrier was created around the sections using an ImmEdge

hydrophobic barrier pen (Cat No. 310018). All incubation steps were carried out at 40 °C using the Pinpoease hybridization system (Cat No. SH-08). The subsequent hybridization, amplification, and detection steps were performed strictly according to the manufacturer's instructions.

Stereotaxic brain surgeries and viral injection

Mice were anesthetized with 1-2% isoflurane during stereotaxic injections, performed using a small animal stereotaxic instrument (RWD Life Science, #68030). Throughout the surgery, core body temperature was maintained at 36 ± 1 °C using a feedback-controlled heating system. Micro scissors were used for the incision, and a dental drill was employed to create the cranial window. Viral injections were delivered using a glass microelectrode syringe pump (RWD Life Science, #R-480). To express GCaMP6m and mCherry AAV2/9-EF1 α -DIO-GCaMP6m-WPRE (titer: 2.89×10^{12} , #H4955, Obio Biotechnology, Shanghai, China, Co., Ltd.) and AAV2/9-EF1 α -DIO-mCherry-WPRE (titer: 2.89×10^{12} , Obio Biotechnology, Shanghai, China, Co., Ltd.) were bilaterally injected into the VMH and rNST, with approximately 0.15 μ L of virus delivered per site. The microelectrode was left in place for 10 minutes post-injection before being withdrawn. Injection coordinates were as follows: VMH (AP, -1.46 mm; ML, ± 0.4 mm; DV, -5.5 mm) and rNST (AP, -7.08 mm; ML, ± 0.6 mm; DV, -4.3 mm). After 3 weeks of AAV injections, optical fibers were chronically implanted in the VMH or rNST and secured with dental cement (C&B Metabond®, Parkell, Japan). Injection coordinates were determined based on the 4th edition of the mouse brain atlas by Franklin and Paxinos (2008). Mice were allowed to recover for 5-7 days post fiber-implanted before any behavioral or calcium imaging evaluations were conducted. After the experiments, mice were perfused to verify virus expression and fiber placement. Data from mice with poor viral expression or incorrect fiber placement (0-20% of cases) were excluded from the final analysis.

For the anterograde tracing experiments, two viral vectors were used. AAV2/1-hsyn-DIO-EGFP-WPRE (titer: 1×10^{13} viral particles/mL, Obio Biotechnology) was injected into the ventromedial hypothalamus (VMH), and AAV2/1-hsyn-SV40 NLS-Cre (titer: 2.1×10^{12} viral particles/mL, Brain Case) was injected into the rostral nucleus of the solitary tract (rNST). Approximately 0.15 μ L of virus was delivered per injection site. After 3 weeks, the brains were perfused and fixed. Subsequently, the brain tissues were sliced and subjected to antibody staining. The fluorescence signals were then recorded and analyzed.

Optical-fiber-based calcium recording in freely mice

Following injection of an AAV2/9-EF1 α -DIO-GCaMP6m-WPRE viral vector, an optical fiber (200 μ m O.D., 0.37 numerical aperture, Inper Inc., China) was placed 150 μ m above the viral injection site. GCaMP6m⁺ mice were implanted with the optic fiber 3 weeks post-AAV injection and allowed to recover for 5-7 days prior to experimental testing.

For **Figure 5K-L** [↗](#), mice were first trained to lick glucose (500 mM) from a petri dish. After training, they were fasted for 12 hours and then received an injection of synthetic NMU peptide, followed by a 30-minute waiting period. During the experimental session, the mice were given 10 seconds to lick glucose from the petri dish while undergoing fiber photometry recording. For **Figure 5E-G** [↗](#), mice were fasted for 4 hours and then administered glucose via gastric infusion, with fiber photometry recordings capturing the process. Fluorescence signals were acquired using a dual-channel fiber photometry system (410 nm & 470 nm, RWD Life Science). The laser power at the tip of the optical fiber was kept below 20 μ W for both the 470 nm and 410 nm channels to minimize bleaching. $\Delta F / F$ was calculated as (470 nm signal - fitted 410 nm signal) / fitted 410 nm signal. Data analysis was performed using software from RWD Life Science.

Ex vivo calcium imaging

For **Figure 1G**, **1I**, **figure supplementary 6**, and **Figure 2**, freshly dissected fly brains were placed in the sugar-free AHL buffer (108 mM NaCl, 8.2 mM MgCl₂, 4 mM NaHCO₃, 1 mM NaH₂PO₄, 5 mM KCl, 2 mM CaCl₂, 5 mM HEPES, pH 7.3). Baseline recordings of the samples in AHL buffer were taken over 1 minutes. Subsequently, the solutions were added with 100 μM glibenclamide (**Figure 2D**), 50 mM pyruvate (**Figure 2G**) or switched to AHL + glucose (50 mM) with or without indicated chemicals (**Figure 2 A and E**, 1 mM phlorizin or 10 μM alloxan), and the pH was adjusted back to 7.3 through gentle perfusion for 3 minutes. Solution flow in the perfusion chamber was controlled by a valve commander (Scientific Instruments). Following stimulation, samples were washed out again with AHL. For the TTX test, 2 μM TTX was added to the AHL solution. Prior to the assay, the samples were pre-treated with 2 mM TTX for 15 minutes.

For **Figure 5H-J**, mice (6-9 weeks of age) were deeply anesthetized by avertin (1.25% avertin, 0.2 ml/10g). Mouse brains were quickly extracted following decapitation and immediately placed in ice-cold slicing buffer (110 mM Choline Cl, 2.5 mM KCl, 1 mM NaH₂PO₄, 25 mM NaHCO₃, 5 mM glucose, 7 mM MgCl₂, 0.5 mM CaCl₂, 1.3 mM Na ascorbate, 0.6 mM Na pyruvate bubbled with 95% oxygen and 5% CO₂ with an adjusted pH of 7.3). The brains were then sectioned into 200 μm slices using a vibratome (Leica VT1200S). These slices were incubated artificial cerebrospinal fluid (aCSF) (125 mM NaCl, 2.5 mM KCl, 1 mM NaH₂PO₄, 25 mM NaHCO₃, 5 mM glucose, 1.3 mM MgCl₂, 2 mM CaCl₂, 1.3 mM Na ascorbate, 0.6 mM Na pyruvate bubbled with 95% oxygen and 5% CO₂ with an adjusted pH of 7.3) at 34 °C for 20 minutes, then transferred to a room temperature for 30 minutes before recording. For Ca²⁺ imaging, the slices were placed in a recording chamber in the low sugar aCSF (125 mM NaCl, 2.5 mM KCl, 1 mM NaH₂PO₄, 25 mM NaHCO₃, 1 mM glucose, 1.3 mM MgCl₂, 2 mM CaCl₂, 1.3 mM Na ascorbate, 0.6 mM Na pyruvate, pH 7.3) for 10 minutes (maintained the flow rate for 1-3 ml/ minute). Baseline recordings, perfusion, and Ca²⁺ recording were performed as detailed above.

All imaging was conducted on an Olympus confocal microscope (FVMPE-RS) with a water immersion objective lens (25× /1.05w MP). Image analyses were performed using ImageJ to calculate the mean intensity of the indicated neuron targeting ROIs, and then plotted in Excel (Microsoft). Ratio changes were calculated using the formula: $\Delta F / F = [F - F_0] / F_0$, where F represents the mean fluorescence of the cell body, and F₀ is the average baseline (before stimulation).

Optogenetics and in vivo calcium imaging

Newly emerged virgin female flies were housed in a fresh vial with standard medium for 3 days and subsequently moved to a vial containing regular food supplemented with 400 μM all-trans-retinal (Sigma R2500) for 2-4 days prior to experiments. These flies were then immobilized on ice, affixed to transparent tape, and the dorsal cuticle of the fly head was delicately removed with forceps to expose the brain, which was immersed in AHL.

For **Figure 3F**, an array of red LED (Thorlabs, M625L3) was positioned above the brain. Calcium signals of Gr5a⁺ neurons were captured for 60 frames (1 second per frame) in the absence of light to establish a baseline. Subsequently, the next 20 frames were recorded during opto-activation with the light (the light was switched on for 0.45 second and rested for 0.55 second per activation cycle, and the images were collected in rest time), followed by another 60 frames recorded without light afterward.

For **Figure 4G** and **4H**, to apply liquid food (5% sucrose) to the proboscis, a pipette was filled with the taste solution and placed a few microns from the proboscis tip by using a micromanipulator (MP225, Sutter Instruments) prior to recording. Then, we recorded calcium signals for 60 seconds in the absence of light to establish a baseline (1 second per frame). To

activate neurons expressing CsChrimson, a red LED was positioned above the brain, controlled by a custom computer program during opto-activation (0.45 second on and 0.55 second off from 60 second to 120 second). The pipette was placed on the proboscis from 80 second to 85 second. Calcium signals of Gr5a⁺ neurons were recorded throughout these processes (1 second per frame).

Image analyses about relative fluorescence changes $\Delta F / F$ were performed as ex vivo calcium imaging.

Optogenetics and ex vivo calcium imaging

For **Figure supplementary 11C** [↗](#), newly emerged virgin female flies were housed in a fresh vial with standard medium for 3 days and subsequently moved to a vial containing regular food supplemented with 400 μ M all-trans-retinal (Sigma R2500) for 2–4 days prior to experiments. The brains were dissected and placed under a red LED. Calcium signals of AstA⁺ neurons were recorded with or without LED light as above. Image analyses were performed as ex vivo calcium imaging.

Calcium imaging with CaMPARI

The flies were then gently immobilized on ice, attached to transparent tape, and the dorsal cuticle of the fly head was carefully removed with forceps to expose the brain, which was immersed in AHL. Photoconversion (PC) was achieved for 30 seconds using a 405 nm LED [Thorlabs, M405L2–UV (405 nm) Mounted LED, 1,000 mA, 410 mW], controlled by a LED controller (Thorlabs, LEDDB1 driven with 1000 mA). Images were captured using an Olympus confocal microscope. Image analyses were performed in ImageJ by manually drawing ROIs covering individual neuronal cell bodies using the green channel. The extent of CaMPARI photoconversion was determined as the Red: Green ratio. A ‘405 nm’ control was included to demonstrate that scanning of the hugin neurons without exposure to ultraviolet light does not convert green to red fluorescence.

Statistical analysis

Data are shown as means ($\pm$ SEM). Data presented in this study were verified for normal distribution by D’Agostino–Pearson omnibus test. Student’s t test, one-way ANOVA and two-way ANOVA (for comparisons among three or more groups and comparisons with more than one variant) were used. The post hoc test with Bonferroni correction was performed for multiple comparisons following ANOVA.

Figure supplements

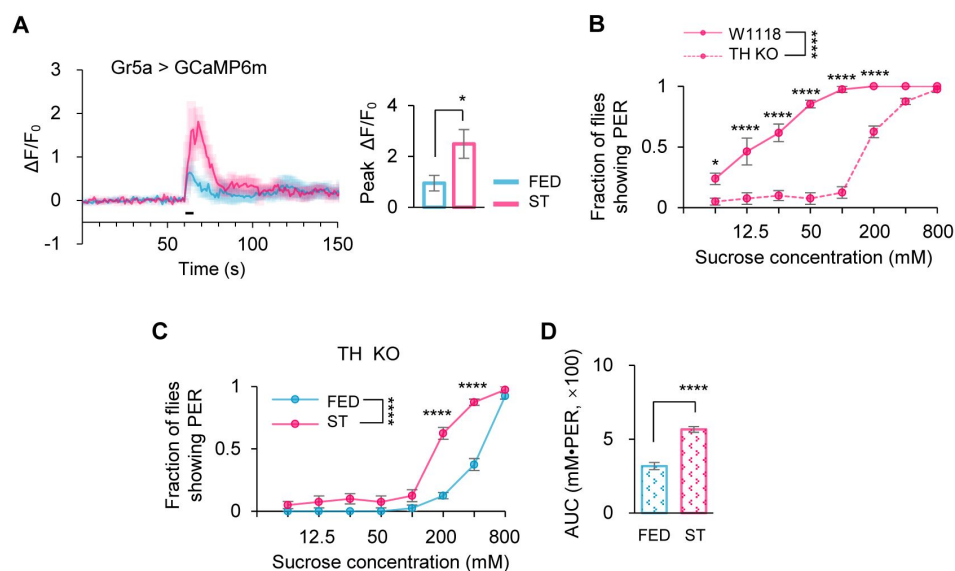


Figure supplementary 1.

Satiety suppressed sweet sensitivity in a dopamine-independent manner.

(A) Representative traces (left) and quantification (right) of peak calcium transients of Gr5a⁺ neurons in indicated flies (n=6). Horizontal black bar represents the duration 5% sucrose stimulation. (B-D) Fraction of flies of the indicated genotypes showing PER to different concentrations of sucrose (B-C, n=4 groups, each of 10 flies). The Area Under the Curve (AUC, D) represents sweet sensitivity in C. *P < 0.05; ****P < 0.0001. Student's *t*-test and two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

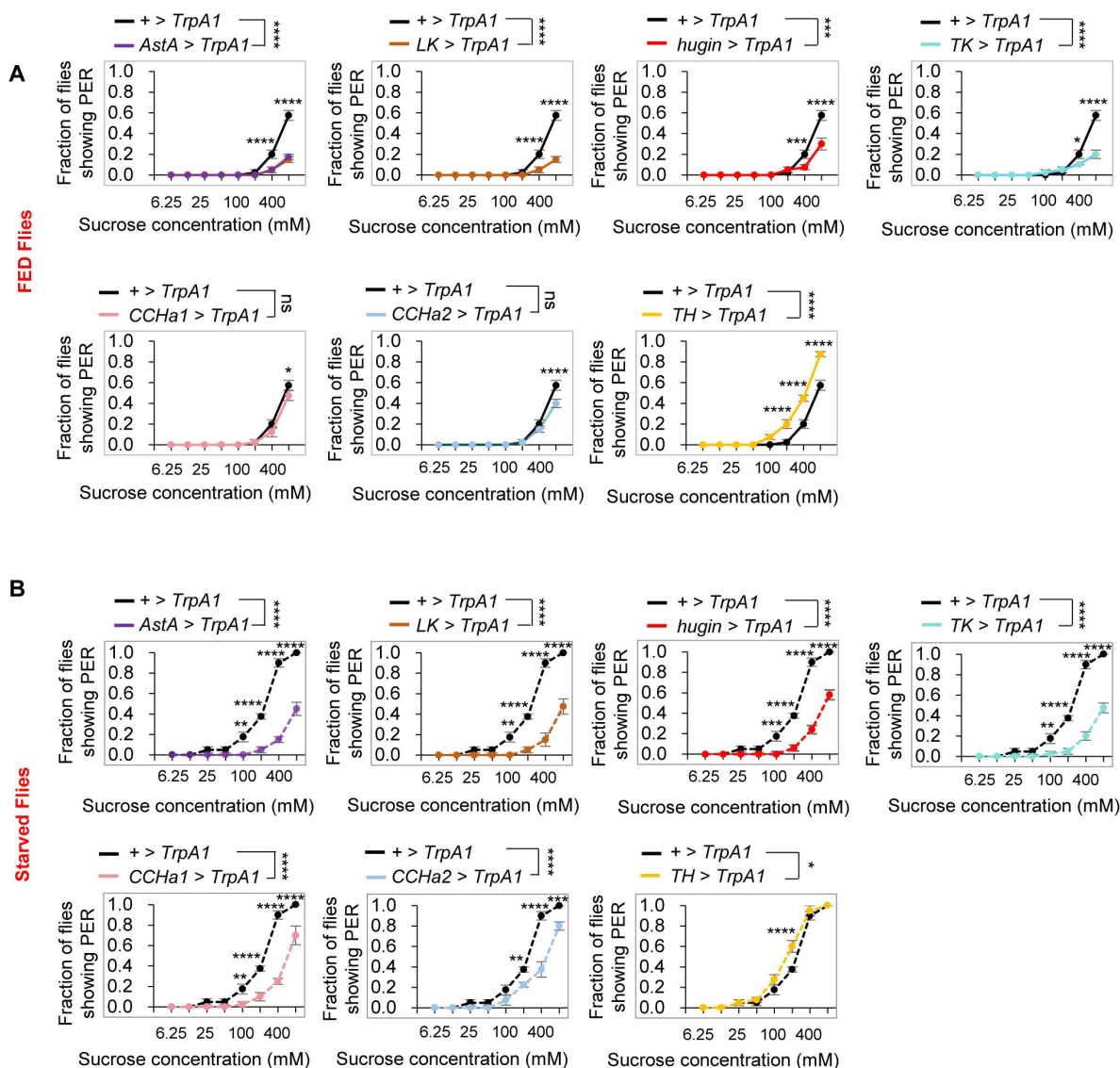


Figure supplementary 2.

The effect of satiety signals on taste modulation.

(A) Fraction of fed flies of the indicated genotypes and environmental temperatures showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). (B) Fraction of starved flies of the indicated genotypes and environmental temperatures showing PER to different concentrations of sucrose (n=4-5 groups, each of 10 flies). ns, $P > 0.05$; * $P < 0.05$; *** $P < 0.001$; **** $P < 0.0001$. Two-way ANOVA followed by post hoc test with Bonferroni correction was used for multiple comparisons when applicable.

Figure supplementary 3.

Activation of IPCs decreased the circulating sugar levels and increased sweet sensitivity.

(A) Hemolymph glucose levels from the indicated genotypes and treatment temperatures (n=8-9, 1 hour at 30 °C). (B-C) Fraction of fed flies of the indicated genotypes and environmental temperatures showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). ns, $P > 0.05$; **** $P < 0.0001$. Student's *t*-test and two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

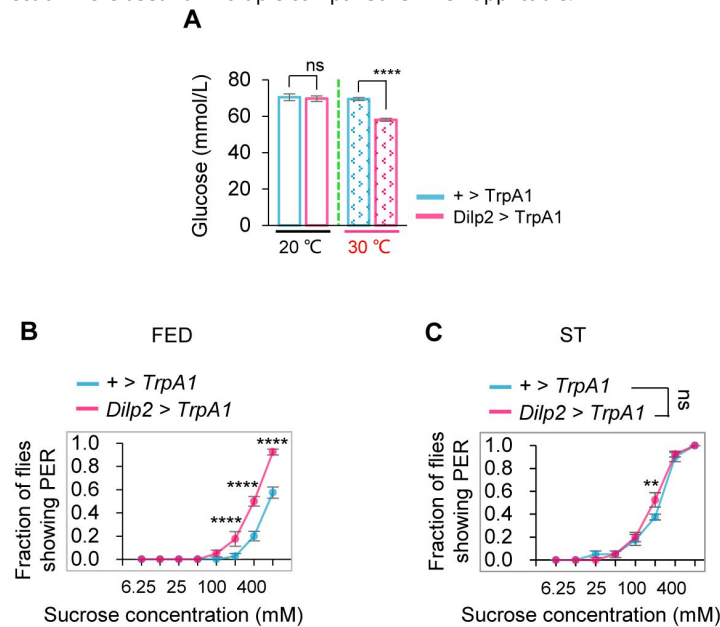


Figure supplementary 4.

Sugar intake suppressed sweet sensation.

(A) Fraction of indicated flies showing PER to different concentrations of fructose (n=4 groups, each of 10 flies). (B) Fraction of indicated flies showing PER to different concentrations of trehalose (n=4 groups, each of 10 flies). **** $P < 0.0001$. two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

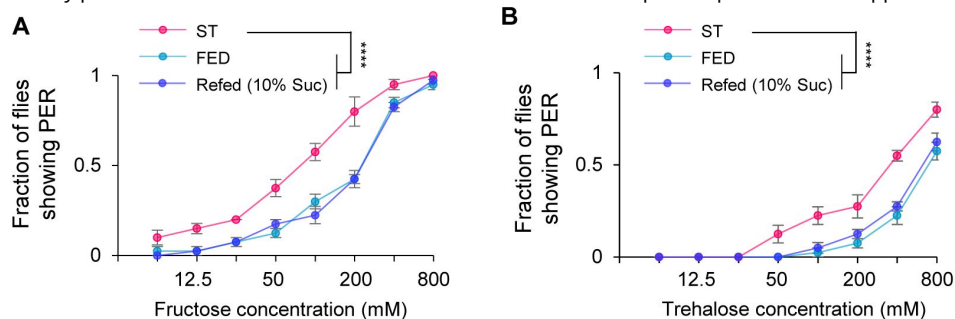


Figure supplementary 5.

Starvation led to a decrease in the circulating sugar levels.

Hemolymph glucose levels from indicated flies (n=6-8). ***P < 0.001. Student's *t*-test was used for comparisons.

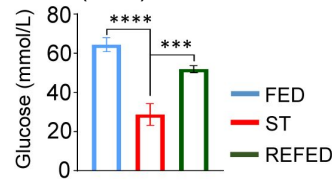
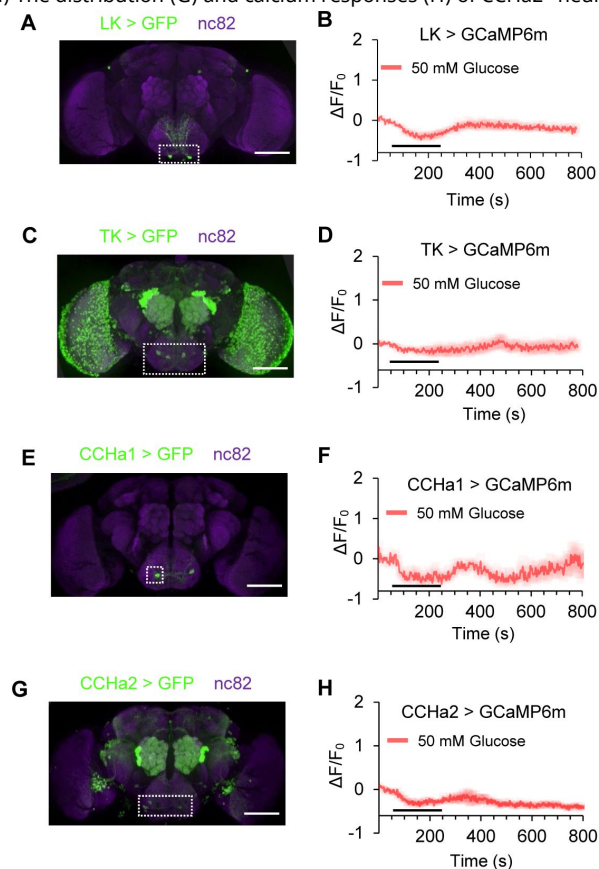


Figure supplementary 6.

The response of different neuropeptide-expressing neurons to glucose.

(A) LK expression in the brain, illustrated by mCD8::GFP expression driven by *LK^{GAL4}*. Scale bar represents 100 μ m. (B) Representative traces and quantification of ex vivo calcium responses of LK⁺ neurons during the perfusion of glucose (n=6). (C-D) The distribution (C) and calcium responses (D) of TK⁺ neurons (n=6). (E-F) The distribution (E) and calcium responses (F) of CCHa1⁺ neurons (n=6). (G-H) The distribution (G) and calcium responses (H) of CCHa2⁺ neurons (n=6).



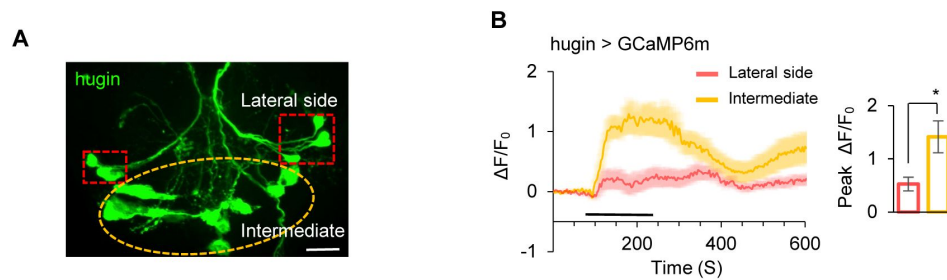


Figure supplementary 7.

The calcium response of hugin neurons upon glucose stimuli.

(A) Representative image of hugin neurons in SEZ. Scale bar represents 50 μm . **(B)** Calcium responses of different cluster of hugin neurons during the perfusion of glucose ($n=6$). * $P < 0.05$; *** $P < 0.001$. Student's *t*-test used for comparisons.

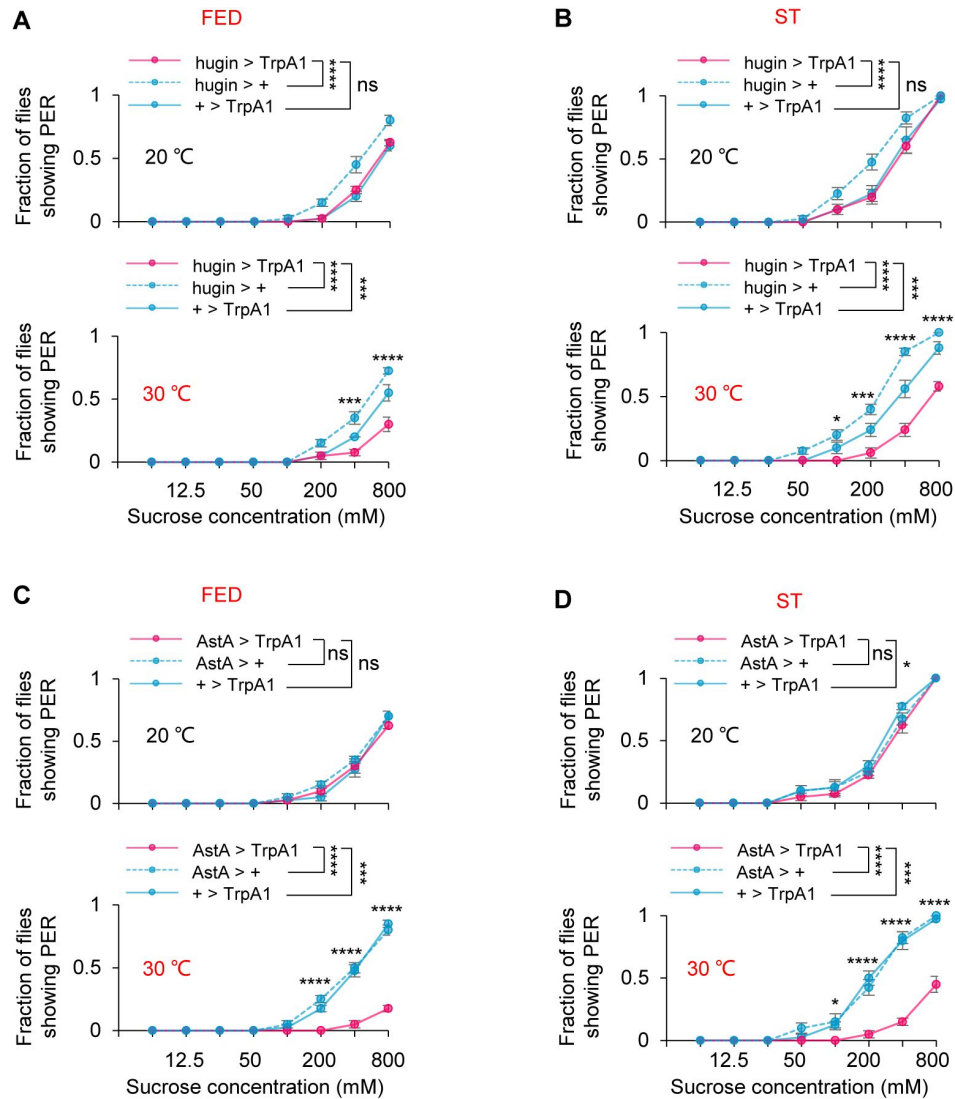


Figure supplementary 8.

hugin⁺ or AstA⁺ neurons suppressed sweet taste.

(A-D) Fraction of flies of the indicated genotypes and environmental temperatures showing PER to different concentrations of sucrose (n=4-5 groups, each of 10 flies). ns, $P > 0.05$; * $P < 0.05$; *** $P < 0.001$; **** $P < 0.0001$. Two-way ANOVA followed by post hoc test with Bonferroni correction was used for multiple comparisons when applicable.

Figure supplementary 9.

The calcium response of *AstA*⁺ neurons upon glucose treatment.

(A) Representative image of different part of *AstA* neurons. Scale bar represents 50 μm . (B) Calcium responses of different cluster of hugin neurons during the perfusion of glucose (n=6)

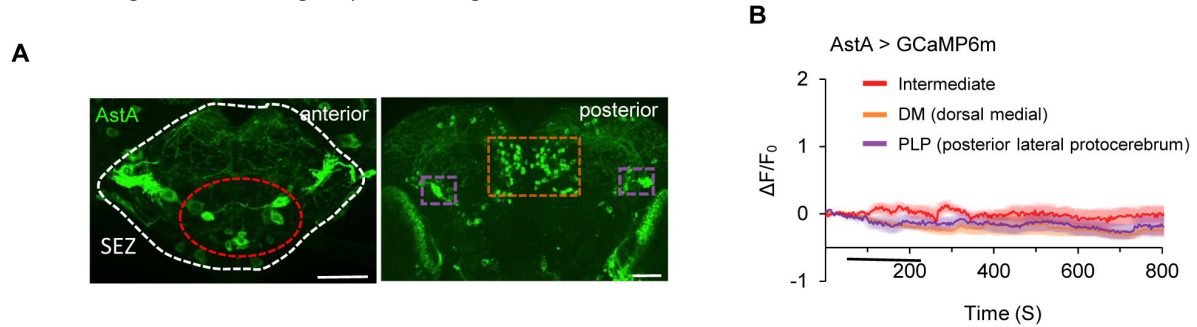


Figure supplementary 10.

Knockout of *PK2-R1* enhanced sweet sensation.

(A-C) Fraction of flies of the indicated genotypes showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). ***P < 0.001; ****P < 0.0001. Two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

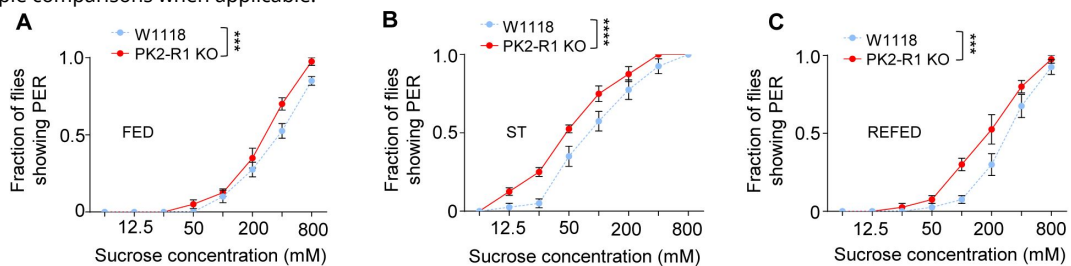


Figure supplementary 11.

hugin⁺ neurons in the brain activated AstA⁺ neurons.

(A-B) Fraction of flies of the indicated genotypes and environmental temperatures showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). Note connection between the brain and VNC was cut off before the behavioral assay. (C) Representative traces (upper) and quantification (lower) of peak calcium transients of AstA⁺ neurons after the photo-activation of hugin⁺ neurons (n=6) from ex vivo calcium imaging. Horizontal black bar represents the duration of red-light stimulation.

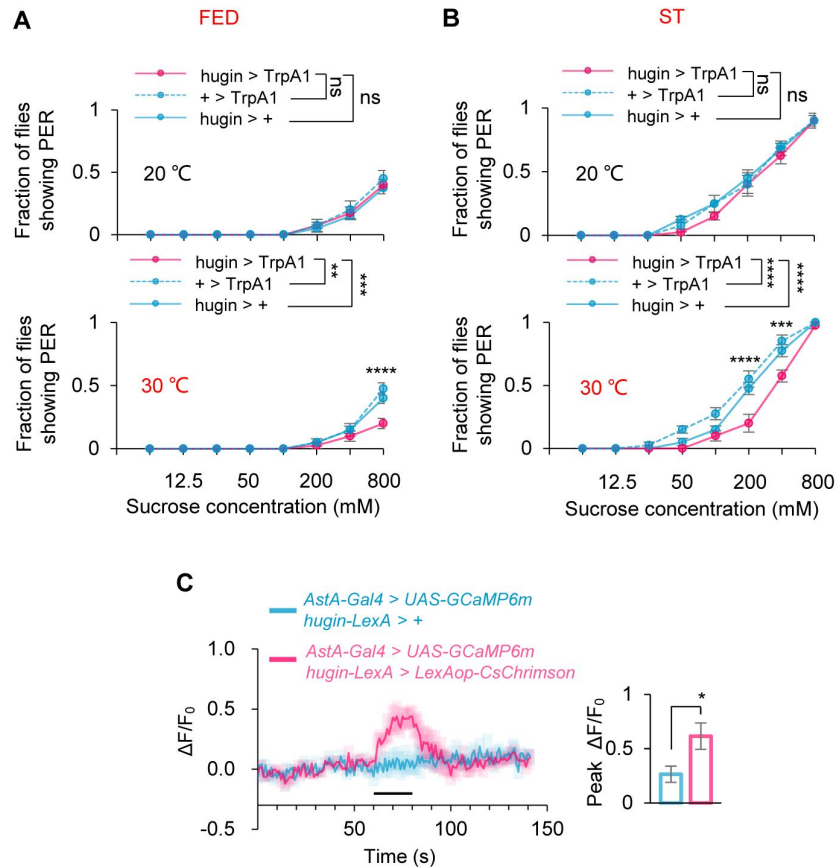
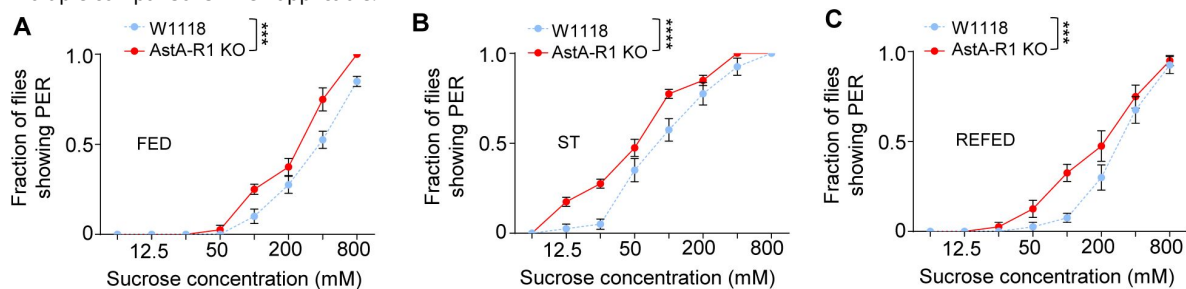


Figure supplementary 12.

Manipulating gene *AstA-R1* enhance PER.

(A-C) Fraction of flies of the indicated genotypes showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). ***P < 0.001; ****P < 0.0001. Two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.



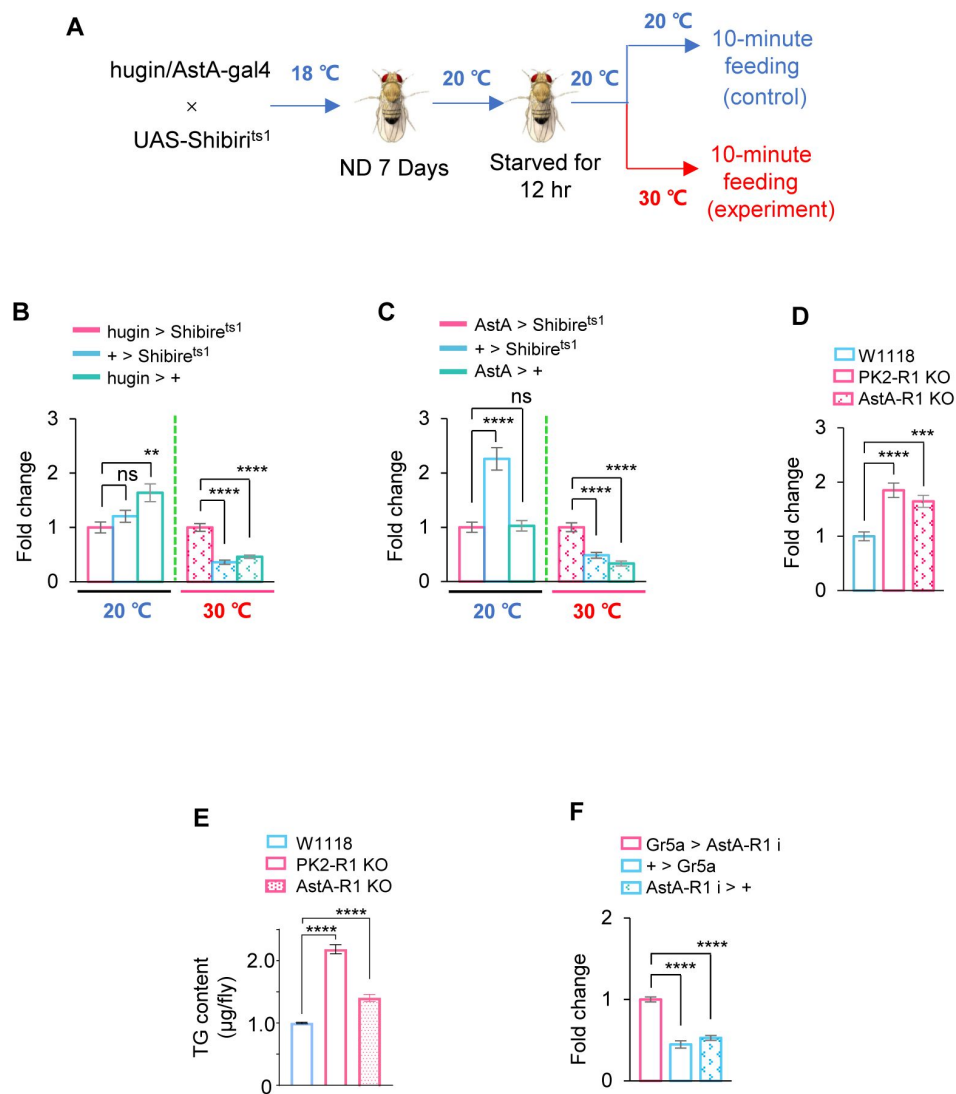


Figure supplementary 13.

hugin-AstA-Gr5a circuitry inhibited feeding behavior.

(A) The illustration of experimental design in [Figure supplementary 13B-C](#). Note that all flies were maintained at 20°C before the feeding assay to prevent neuronal silencing by Shibire^{ts1} during these procedures. (B-C) Food consumption of flies of the indicated genotypes and environmental temperatures (n=8 groups, each of 4 flies). (D) Food consumption of flies of the indicated genotypes (n=8 groups, each of 4 flies). (E) The TG content of indicated genotypes (n=8 groups, each of 2 flies). (F) Food consumption of flies of the indicated genotypes (n=8 groups, each of 4 flies). ns, $P > 0.05$; *** $P < 0.001$; **** $P < 0.0001$. Student's *t*-test and two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

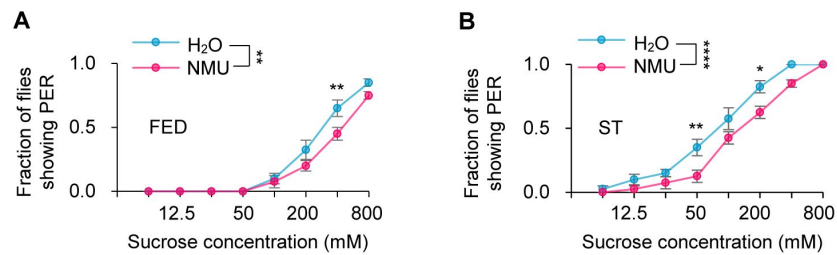


Figure supplementary 14.

NMU peptide suppressed sweet taste in fly.

Fraction of flies showing PER to different concentrations of sucrose (n=4 groups, each of 10 flies). Flies were injected with saline or synthetic NMU for 30 mins before the assay. **P < 0.01; ****P < 0.0001. Two-way ANOVA followed by post hoc test with Bonferroni correction were used for multiple comparisons when applicable.

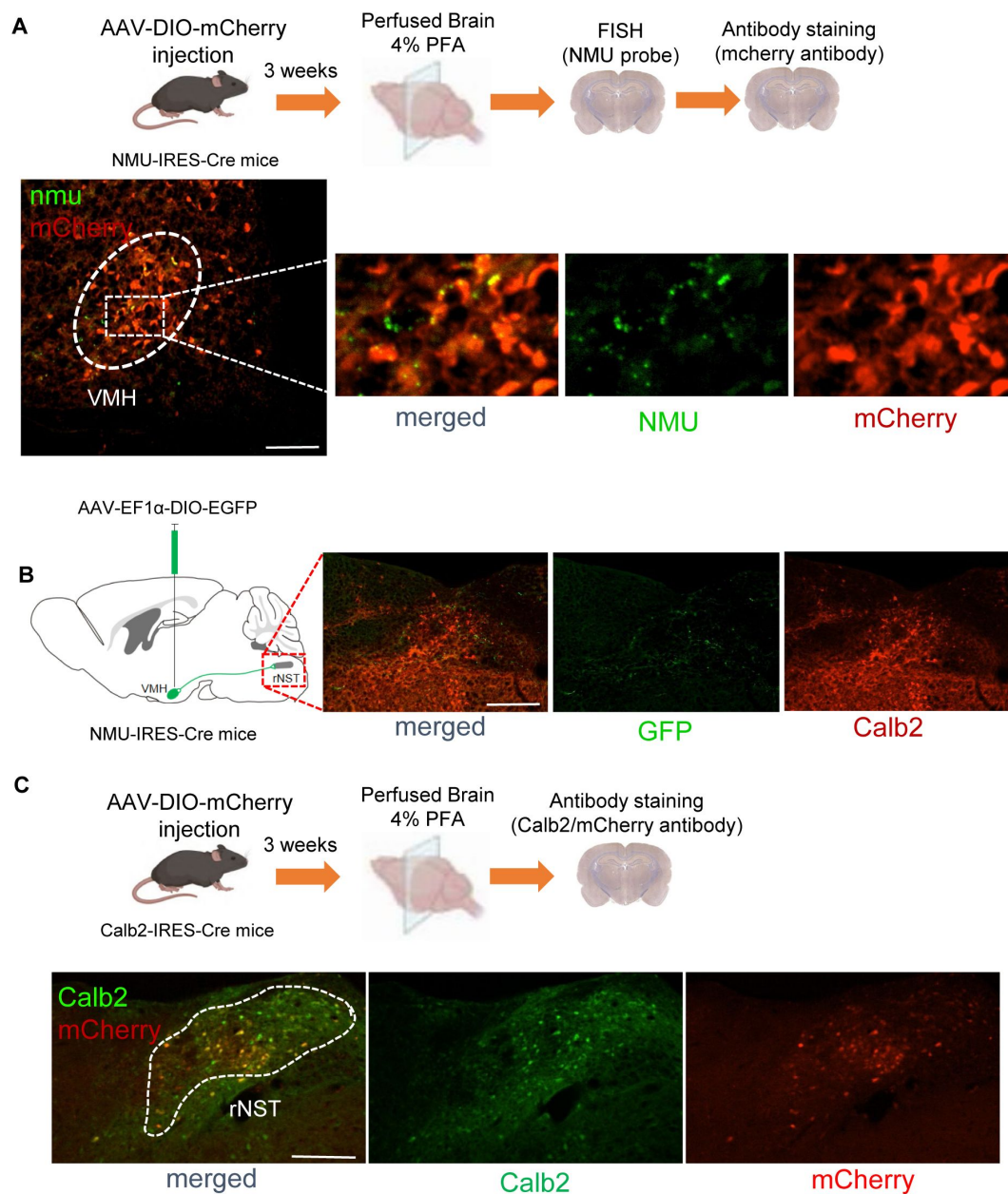


Figure supplementary 15.

The expressing pattern of NMU neurons and Calb2 neurons

(A) Co-labeling of NMU⁺ neurons using a Cre-dependent mCherry reporter (red) and in situ hybridization (green). Scale bar represents 200 μ m. (B) NMU arbors was extended near Calb2^{rNST} neurons. Scale bar represents 200 μ m. (C) Co-labeling of Calb2⁺ neurons with a Cre-dependent mCherry reporter (red) and NMU antibody staining (green). Scale bar represents 200 μ m.

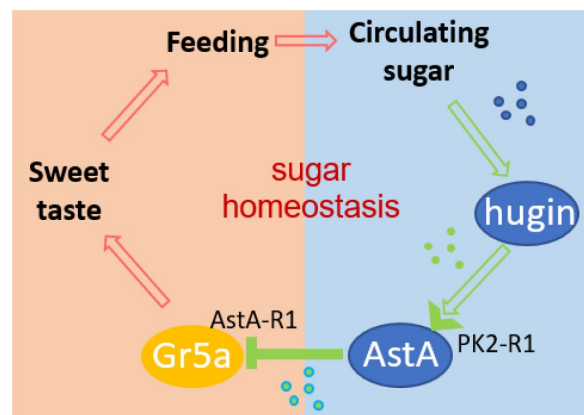


Figure supplementary 16.

A working model

In the fly brain, a small group of neurons expressing hugin peptide is responsible for detecting the circulating glucose levels. Following feeding, as the levels of glucose in the circulatory system increased, these neurons are activated, leading to the release of hugin, which activated AstA neurons via its receptor PK2-R1. Subsequently, the activation of AstA⁺ neurons then directly inhibited sweet perception via AstA peptide and its cognate receptor AstA-R1 expressed in sweet-sensing Gr5a⁺ neurons. This neural circuitry is a novel energy sensor that suppressed sweet perception and terminated feeding behavior, providing an efficient mechanism for maintaining energy homeostasis.

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Additional information

Author contributions

T.S., R.H., and L.W. designed the experiments and wrote the manuscript. W.Q., and T.S. conducted and analyzed all the experiments. L.W. and R.H. supervised the project.

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Reviewer #1 (Public review):

In this manuscript, Qin and colleagues aim to delineate a neural mechanism by which the internal satiety levels modulate the intake of sugar solution. They identified a three-step neuropeptidergic system that downregulates the sensitivity of sweet-sensing gustatory sensory neurons in sated flies. First, neurons that release a neuropeptide Hugin (which is an insect homolog of vertebrate Neuromedin U (NMU)) are in an active state when the concentration of glucose is high. This activation does not require synaptic inputs, suggesting that Hugin-releasing neurons sense hemolymph glucose levels directly. Next, the Hugin neuropeptides activate Allatostatin A (AstA)-releasing neurons via one of Hugin's receptors, PK2-R1. Finally, the released AstA neuropeptide suppresses sugar response in sugar-sensing Gr5a-expressing gustatory sensory neurons through AstA-R1 receptor. Suppression of sugar response in Gr5a-expressing neurons reduces the fly's sugar intake motivation (measured by proboscis extension reflex). They also found that NMU-expressing neurons in the ventromedial hypothalamus (VMH) of mice (which project to the rostral nucleus of the

solitary tract (rNST)) are also activated by high concentrations of glucose, independent of synaptic transmission, and that injection of NMU reduces the glucose-induced activity in the downstream of NMU-expressing neurons in rNST. These data suggest that the function of Hugin neuropeptide in the fly is analogous to the function of NMU in the mouse.

Generally, their central conclusions are well-supported by multiple independent approaches. The parallel study in mice adds a unique comparative perspective that makes the paper interesting to a wide range of readers. It is easier said than done: the rigor of this study, which effectively combined pharmacological and genetic approaches to provide multiple lines of behavioral and physiological evidence, deserves recognition and praise.

A perceived weakness is that the behavioral effects of the manipulations of Hugin and AstA systems are modest compared to a dramatic shift of sugar solution-induced PER (the behavioral proxy of sugar sensitivity) induced by hunger, as presented in Figure 1B and E. It is true that the mutation of tyrosine hydroxylase (TH), which synthesizes dopamine, does not completely abolish the hunger-induced PER change, but the remaining effect is small. Moreover, the behavioral effect of the silencing of the Hugin/AstA system (Figure Supplement 13B, C) is difficult to interpret, leaving a possibility that this system may not be necessary for shifting PER in starved flies. These suggest that the Hugin-AstA system accounts for only a minor part of the behavioral adaptation induced by the decreased sugar levels. Their aim to "dissect out a complete neural pathway that directly senses internal energy state and modulates food-related behavioral output in the fly brain" is likely only partially achieved. While this outcome is not a shortcoming of a study per se, the depth of discussion on the mechanism of interactions between the Hugin/AstA system and the other previously characterized molecular circuit mechanisms mediating hunger-induced behavioral modulation is insufficient for readers to appreciate the novelty of this study and future challenges in the field. In this context, authors are encouraged to confront a limitation of the study due to the lack of subtype-level circuit characterization, despite their intriguing finding that only a subtype of Hugin- and AstA-releasing neurons are responsive to the elevated level of bath-applied glucose.

<https://doi.org/10.7554/eLife.108551.1.sa3>

Reviewer #2 (Public review):

Summary:

The question of how caloric and taste information interact and consolidate remains both active and highly relevant to human health and cognition. The authors of this work sought to understand how nutrient sensing of glucose modulates sweet sensation. They found that glucose intake activates hugin signaling to AstA neurons to suppress feeding, which contributes to our mechanistic understanding of nutrient sensation. They did this by leveraging the genetic tools of *Drosophila* to carry out nuanced experimental manipulations and confirmed the conservation of their main mechanism in a mammalian model. This work builds on previous studies examining sugar taste and caloric sensing, enhancing the resolution of our understanding.

Strengths:

Fully discovering neural circuits that connect body state with perception remains central to understanding homeostasis and behavior. This study expands our understanding of sugar sensing, providing mechanistic evidence for a hugin/AstA circuit that is responsive to sugar intake and suppresses feeding. In addition to effectively leveraging the genetic tools of *Drosophila*, this study further extends their findings into a mammalian model with the discovery that NMU neural signaling is also responsive to sugar intake.

Weaknesses:

The effect of Glut1 knockdown on PER in hugin neurons is modest, and does not show a clear difference between fed and starved flies as might be expected if this mechanism acts as a sensor of internal energy state. This could suggest that glucose intake through Glut1 may only be part of the mechanism.

<https://doi.org/10.7554/eLife.108551.1.sa2>

Reviewer #3 (Public review):**Summary:**

This study identifies a novel energy-sensing circuit in *Drosophila* and mice that directly regulates sweet taste perception. In flies, hugin⁺ neurons function as a glucose sensor, activated through Glut1 transport and ATP-sensitive potassium channels. Once activated, hugin neurons release hugin peptide, which stimulates downstream Allatostatin A (AstA)⁺ neurons via PK2-R1 receptors. AstA⁺ neurons then inhibit sweet-sensing Gr5a⁺ gustatory neurons through AstA peptide and its receptor AstA-R1, reducing sweet sensitivity after feeding. Disrupting this pathway enhances sweet taste and increases food intake, while activating the pathway suppresses feeding.

The mammalian homolog of neuromedin U (NMU) was shown to play an analogous role in mice. NMU knockout mice displayed heightened sweet preference, while NMU administration suppressed it. In addition, VMH NMU⁺ neurons directly sense glucose and project to rNST Calb2⁺ neurons, dampening sweet taste responses. The authors suggested a conserved hugin/NMU-AstA pathway that couples energy state to taste perception.

Strengths:

Interesting findings that extend from insects to mammals. Very comprehensive.

Weaknesses:

Coupling energy status to taste sensitivity is not a new story. Many pathways appear to be involved, and therefore, it raises a question as to how this hugin-AstA pathway is unique.

<https://doi.org/10.7554/eLife.108551.1.sa1>

Author response:**Reviewer #1 (Public review):**

In this manuscript, Qin and colleagues aim to delineate a neural mechanism by which the internal satiety levels modulate the intake of sugar solution. They identified a three-step neuropeptidergic system that downregulates the sensitivity of sweet-sensing gustatory sensory neurons in sated flies. First, neurons that release a neuropeptide Hugin (which is an insect homolog of vertebrate Neuromedin U (NMU)) are in an active state when the concentration of glucose is high. This activation does not require synaptic inputs, suggesting that Hugin-releasing neurons sense hemolymph glucose levels directly. Next, the Hugin neuropeptides activate Allatostatin A (AstA)-releasing neurons via one of Hugin's receptors, PK2-R1. Finally, the released AstA neuropeptide suppresses sugar response in sugar-sensing Gr5a-expressing gustatory sensory neurons through AstA-R1 receptor. Suppression of sugar response in Gr5a-expressing neurons reduces the

fly's sugar intake motivation (measured by proboscis extension reflex). They also found that NMU-expressing neurons in the ventromedial hypothalamus (VMH) of mice (which project to the rostral nucleus of the solitary tract (rNST)) are also activated by high concentrations of glucose, independent of synaptic transmission, and that injection of NMU reduces the glucose-induced activity in the downstream of NMU-expressing neurons in rNST. These data suggest that the function of Hugin neuropeptide in the fly is analogous to the function of NMU in the mouse.

Generally, their central conclusions are well-supported by multiple independent approaches. The parallel study in mice adds a unique comparative perspective that makes the paper interesting to a wide range of readers. It is easier said than done: the rigor of this study, which effectively combined pharmacological and genetic approaches to provide multiple lines of behavioral and physiological evidence, deserves recognition and praise.

A perceived weakness is that the behavioral effects of the manipulations of Hugin and AstA systems are modest compared to a dramatic shift of sugar solution-induced PER (the behavioral proxy of sugar sensitivity) induced by hunger, as presented in Figure 1B and E. It is true that the mutation of tyrosine hydroxylase (TH), which synthesizes dopamine, does not completely abolish the hunger-induced PER change, but the remaining effect is small. Moreover, the behavioral effect of the silencing of the Hugin/AstA system (Figure Supplement 13B, C) is difficult to interpret, leaving a possibility that this system may not be necessary for shifting PER in starved flies. These suggest that the Hugin-AstA system accounts for only a minor part of the behavioral adaptation induced by the decreased sugar levels. Their aim to "dissect out a complete neural pathway that directly senses internal energy state and modulates food-related behavioral output in the fly brain" is likely only partially achieved. While this outcome is not a shortcoming of a study per se, the depth of discussion on the mechanism of interactions between the Hugin/AstA system and the other previously characterized molecular circuit mechanisms mediating hunger-induced behavioral modulation is insufficient for readers to appreciate the novelty of this study and future challenges in the field.

We thank the reviewer for the thoughtful comment. We agree that the behavioral effects of manipulating the Hugin–AstA system alone were considerably weaker than the pronounced PER shifts induced by starvation. We will revise our Discussion to address it by positioning our findings within the broader context of energy regulation.

More specifically, we will discuss that feeding behavior is controlled by two distinct, yet synergistic, types of mechanisms:

- (1) Hunger-driven 'accelerators': as the reviewer notes, pathways involving dopamine and NPF are powerful drivers of sweet sensitivity. These systems are strongly activated by hunger to promote food-seeking and consumption.
- (2) Satiety-driven 'brakes': our study identifies the counterpart to those systems above, aka. a satiety-driven 'brake'. The Hugin–AstA pathway acts as a direct sensor of high internal energy (glucose), which is specifically engaged during satiety to actively suppress sweet sensation and prevent overconsumption.

This framework explains the seemingly discrepancy in effect size. The dramatic PER shift seen upon starvation is a combined result of engaging the 'accelerators' (hunger pathways like TH/NPF) while simultaneously releasing the 'brake' (our Hugin–AstA pathway being inactive).

Our manipulations, which specifically target only the 'brake' system, are therefore expected to have a more modest effect than this combined physiological state. Thus, rather than being a "minor part," the Hugin–AstA pathway is a mechanistically defined, satiety-specific circuit that is essential for the precise "braking" required for energy homeostasis. We will update our Discussion to emphasize how these 'accelerator' and 'brake' circuits must work in concert to ensure precise energy regulation.

In this context, authors are encouraged to confront a limitation of the study due to the lack of subtype-level circuit characterization, despite their intriguing finding that only a subtype of Hugin- and AstA-releasing neurons are responsive to the elevated level of bath-applied glucose.

We thank the reviewer for highlighting the critical issue of subtype-level specialization within the Hugin and AstA populations.

We fully agree that the Hugin system is known for its functional heterogeneity (pleiotropy), with different Hugin neuron subclusters implicated in regulating a variety of behaviors, including feeding, aversion, and locomotion (we will cite relevant literature here). Our finding that only a specific subcluster of Hugin neurons is responsive to glucose elevation provides a crucial first step in functionally dissecting this complexity.

We will add a dedicated paragraph to elaborate on this functional partitioning. We propose that this subtype-level specialization allows the Hugin system to precisely link specific physiological states (like high circulating glucose) to appropriate behavioral outputs (like the suppression of sweet taste), demonstrating an elegant solution to coordinating multiple survival behaviors. Future work using high-resolution tools such as split-GAL4 and single-cell sequencing will be invaluable in fully mapping the specific functional roles corresponding to each Hugin and AstA subcluster.

Reviewer #2 (Public review):

Summary:

The question of how caloric and taste information interact and consolidate remains both active and highly relevant to human health and cognition. The authors of this work sought to understand how nutrient sensing of glucose modulates sweet sensation. They found that glucose intake activates hugin signaling to AstA neurons to suppress feeding, which contributes to our mechanistic understanding of nutrient sensation. They did this by leveraging the genetic tools of Drosophila to carry out nuanced experimental manipulations and confirmed the conservation of their main mechanism in a mammalian model. This work builds on previous studies examining sugar taste and caloric sensing, enhancing the resolution of our understanding.

Strengths:

Fully discovering neural circuits that connect body state with perception remains central to understanding homeostasis and behavior. This study expands our understanding of sugar sensing, providing mechanistic evidence for a hugin/AstA circuit that is responsive to sugar intake and suppresses feeding. In addition to effectively leveraging the genetic tools of Drosophila, this study further extends their findings into a mammalian model with the discovery that NMU neural signaling is also responsive to sugar intake.

Weaknesses:

The effect of Glut1 knockdown on PER in hugin neurons is modest, and does not show a clear difference between fed and starved flies as might be expected if this mechanism

acts as a sensor of internal energy state. This could suggest that glucose intake through Glut1 may only be part of the mechanism.

We thank the reviewer for this insightful comment and agree that the modest behavioral effect of Glut1 knockdown is a critical finding that warrants further clarification. This observation strongly supports the idea that internal energy state is monitored by a sophisticated and robust network, not a single, fragile component. We believe the effect size is modest for two main reasons, which we will further address in revised Discussion.

Firstly, the effect size is likely attenuated by technical and molecular redundancy. Specifically, the RNAi-mediated knockdown of Glut1 may be incomplete, leaving residual transporter function. Furthermore, Glut1 is likely only one part of the Hugin neuron's intrinsic sensing mechanism; other components, such as alternative glucose transporters or downstream K_{ATP} channel signaling, may provide molecular redundancy, meaning that the full energy-sensing function is not easily abolished by a single manipulation.

Secondly, and more importantly, the final feeding decision is an integrated output of competing circuits. While hunger-sensing pathways like the dopamine and NPF circuits act as powerful "accelerators" to drive sweet consumption, the Hugin–AstA pathway serves as a satiety-specific "brake". The modest effect of partially inhibiting just one component of this 'brake' system is the hallmark of a precisely regulated, multi-layered homeostatic system. We will further clarify in the Discussion that the Hugin pathway represents one essential inhibitory circuit within this cooperative network that works together with the hunger-promoting systems to ensure precise control over energy intake.

Reviewer #3 (Public review):

Summary:

This study identifies a novel energy-sensing circuit in Drosophila and mice that directly regulates sweet taste perception. In flies, hugin+ neurons function as a glucose sensor, activated through Glut1 transport and ATP-sensitive potassium channels. Once activated, hugin neurons release hugin peptide, which stimulates downstream Allatostatin A (AstA)+ neurons via PK2-R1 receptors. AstA+ neurons then inhibit sweet-sensing Gr5a+ gustatory neurons through AstA peptide and its receptor AstA-R1, reducing sweet sensitivity after feeding. Disrupting this pathway enhances sweet taste and increases food intake, while activating the pathway suppresses feeding.

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Strengths

Interesting findings that extend from insects to mammals. Very comprehensive.

Weaknesses:

Coupling energy status to taste sensitivity is not a new story. Many pathways appear to be involved, and therefore, it raises a question as to how this hugin-AstA pathway is unique.

The reviewer is correct that several energy-sensing pathways are known. However, we now clarify that these previously established mechanisms, such as the dopaminergic and NPF

pathways, primarily function as hunger-driven "accelerators." They are activated by low energy states to promote sweet sensitivity and drive consumption.

The crucial, missing piece of the puzzle—which our study provides—is the satiety-specific "brake" mechanism. We identify the Hugin–AstA circuit as one of the "brakes": a dedicated, central sensor that responds directly to high circulating glucose (satiety) to suppress sweet sensation and prevent overconsumption.

Thus, our work is unique because it defines the essential counterpart to the hunger pathways. In the revised Discussion, we will further explain how these 'accelerator' (hunger) and 'brake' (satiety) systems work in concert to allow for the precise, bidirectional regulation of energy intake. Furthermore, by demonstrating that this Hugin/NMU 'brake' circuit is evolutionarily conserved in mice, our findings reveal a fundamental energy-sensing strategy and suggest that this pathway could represent a promising new therapeutic target for managing conditions of excessive food intake.

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